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Evaluating and Operationalising XAI Methods for ML Energy Commodities Forecasting

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This dissertation is submitted in partial fulfilment of the requirements of the degree of MSc Computational Finance at UCL. It is substantially the result of my own work except where explicitly indicated in the text. The dissertation may be freely copied and distributed provided the source is explicitly acknowledged.

Abstract

This thesis is an applied investigation of explainable AI (XAI) methods for machine learning (ML) energy commodities forecasting models. XAI methods have emerged as a leading paradigm for post-hoc interpretability, addressing concerns around black-box models in systematic trading. However, their application to autocorrelated financial time series remains underexplored, and their potential to serve as real-time confidence signals is an open research gap. The core objectives of this thesis are: (1) to evaluate how recent XAI developments compare to established methods on financial time-series, (2) to identify patterns in feature attributions across XAI methods and ML models, and (3) to operationalise these signals to enhance trading performance.

Three ML-based models from distinct modelling paradigms are built: an **XGBoost** model with no temporal structure, a Long Short-Term Memory (**LSTM**) network that captures temporal dependencies, and **TabPFN-3** – a transformer-based in-context learning foundation model pre-trained on synthetic tabular datasets via prior-fitted networks. This diversity allows for examination of whether XAI explanations vary across fundamentally different model types. The models target weekly returns of the Bloomberg Energy Subindex (BCOMENTR), and are trained on 2004 – 2026 macroeconomic, market, and energy-specific data. Three XAI methods are implemented across the models: **SHAP**, **LIME**, and **WinIT** – enabling comparison of fundamentally different approaches to measuring feature importance.

The investigation is structured around 3 experiments:

Exp. 1 – Comparison of Explanations Across Methods and Models: an analysis of global feature importance across XAI methods and models. Cross-method and cross-model consistency is quantified using rank correlation. Faithfulness is evaluated via faithfulness correlation to test whether XAI-attributed importance corresponds to actual predictive contribution. Stability is assessed using Gaussian-perturbed inputs, producing a quantitative robustness score per method-model pair. This experiment verifies which XAI methods produce trustworthy explanations.

Exp. 2 – Local Attribution Patterns Across Prediction Outcomes: XAI explanations are computed for each out-of-sample prediction and partitioned by directional accuracy. This experiment identifies whether systematic shifts in feature importance or temporal lag structure are detectable during model errors. Of particular interest is whether cross-method agreement degrades or LSTM models exhibit unrealistic temporal attribution profiles during incorrect predictions.

Exp. 3 – XAI-Derived Trading Signal: the positive results of Experiment 2 are operationalised into a novel XAI-derived confidence-gating mechanism. For each prediction, the local attribution vector is compared against the characteristic error-state profiles identified in Experiment 2. Performance is backtested on a held-out test-

set (January 2024 onwards) and benchmarked against an all-predictions strategy and a buy-and-hold baseline. This potentially constitutes the first empirical application of XAI attribution patterns as a real-time confidence gate – a research direction inspired by recent literature.

This thesis presents the following original contributions to science:

- 1) **Novel XAI-derived confidence-gating mechanism for algorithmic trading:** This thesis introduced a method for converting local XAI attribution patterns into a real-time confidence signal. This operationalisation of XAI beyond simple post-hoc interpretation is a novel contribution not previously tested in the literature.
- 2) **Established XAI faithfulness and stability analysis applied in a new domain:** This work adds to existing XAI evaluation literature by applying established faithfulness and stability metrics in the specific domain of energy commodities time series.
- 3) **Characterisation of local XAI attribution shifts under model error:** The study identifies systematic differences in local feature attribution patterns between correct and incorrect predictions, contributing empirical evidence that explanation characteristics are correlated with prediction unreliability in financial time-series.

Additionally, this thesis presents the following contributions to industry:

- 1) **Practical confidence-gating framework compatible with existing systematic strategies:** The XAI-derived confidence signal is designed to operate as an overlay on any trading strategy, requiring no model re-specification. This makes it directly deployable within existing trading systems, with the potential to improve risk-adjusted performance without altering the underlying alpha model.
- 2) **Feature importance rankings for energy commodities trading desks:** The global and local XAI analysis produces financially interpretable feature importance rankings for energy commodities forecasting, identifying which macro, market, inventory, and cross-asset features can provide valuable trading signals.
- 3) **XAI method selection guidance for practitioners:** The faithfulness and stability analysis provides empirical guidance on which XAI methods are most reliable for financial time-series and different ML models. This addresses a practical need of quant teams integrating XAI into model validation workflows.

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Chapter 1

Introduction

This chapter presents an overview of the thesis. It first introduces and motivates the domain of ML forecasting models in commodities and finance in general, and the problem of interpreting and explaining these models which XAI methods attempt to solve. It identifies the key gaps in existing XAI literature which this thesis aims to explore. This leads on to the primary research objectives, followed by a summary of the 3 core experiments, and a description of the key scientific and industry contributions. It concludes with an outline of the thesis structure.

1.1. Research Motivation

Algorithmic trading has come to dominate modern financial markets, driven by the speed and scalability advantages of automated execution over manual decision-making. High-frequency trading has demonstrated the potential to exploit these advantages for excess returns. As trading infrastructure has matured and computational capacity has increased, the techniques utilised within these algorithmic trading systems have evolved substantially: while early approaches primarily relied on technical indicators and rule-based logic, the field is increasingly adopting machine learning (ML) and deep learning (DL) techniques.

The adoption of ML models has been driven primarily by two capabilities. Firstly, ML models can recognise complex, non-linear patterns in large, high-dimensional datasets that traditional statistical or technical approaches struggle to capture. Secondly, modern strategies often incorporate richer datasets beyond price history alone, such as economic indicators, sentiment data, and other alternative datasets – allowing models to exploit a deeper information set than purely technical strategies can. The result has been a rapid proliferation of ML and DL techniques across the trading industry as practitioners seek to extract an advantage from increasingly complex, data-rich environments.

However, this proliferation has surfaced a fundamental issue: the black-box nature of the DL architectures and ensemble methods being used. Their internal decision-making processes are largely opaque to the practitioners who deploy them. In high-stakes financial decision-making, where model outputs directly inform capital allocation and risk exposure, the inability to understand why a model produces a given prediction creates significant operational and governance risks. Regulatory bodies have directly responded to the growing use of opaque AI systems in finance. For example, the EU AI Act classifies AI systems used in critical financial decision-making as high-risk, mandating greater transparency, documentation, and human oversight. Model opacity is no longer simply a technical inconvenience, but increasingly a compliance and

regulatory requirement that institutions adopting ML-based trading systems must directly address.

Explainable AI (XAI) has emerged over the last decade as the leading paradigm for addressing this opacity, by generating post-hoc explanations of black-box model predictions. The most popular XAI methods are SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), which offer global insight into overall model behaviour as well as local insight into individual predictions. Their strengths and limitations have been rigorously tested within the literature, with a set of standard formal evaluation criteria having been developed for benchmarking XAI methods. This includes faithfulness (whether explanations accurately reflect predictive performance contribution) and stability (the extent to which small random perturbations to the input data affect explanations).

Despite the field of XAI being well established, analysis of XAI applied to financial time series specifically remains a relatively underdeveloped subfield. Financial time series exhibits numerous unique characteristics, ranging from low-signal-to-noise ratios, to non-stationarity, to the key challenge of sequential, temporal data where observations cannot be assumed to be independently and identically distributed across the time dimension. Existing research has demonstrated that key XAI method characteristics like faithfulness can differ significantly across domains and datasets. Mercier, et al. (2022), in a comprehensive benchmark of time series XAI methods, found that explanation faithfulness (measured via infidelity) varies substantially across datasets even within the time series domain. In fact, no single XAI method or method category delivered consistently superior performance across datasets. Therefore, there is a clear need to evaluate XAI methods across different domains, such as on financial time series, and on specific asset classes with their own distinct volatility and structural dynamics. Without such testing, conclusions about XAI faithfulness and stability drawn from other domains cannot simply be assumed to transfer to all financial forecasting applications.

The challenge of sequential data has also led to the recent development of time-series specific XAI methods such as WinIT and Dynamask, which account for temporal dependency between observations. These newer methods have not yet received the extensive empirical scrutiny applied to their more established counterparts, leaving open questions about how well they perform in practice.

Beyond these areas of comparatively limited research, more fundamental gaps remain. First, the existing literature applying XAI to financial time series has disproportionately focused on relatively simple model architectures, particularly LSTMs and other standard neural networks. Meanwhile, the ML field has advanced significantly, with development of state of the art models such as self-learning agents and pre-trained foundation models capable of in-context learning across diverse tasks. Understanding how XAI methods perform when applied to these newer architectures has therefore become a pressing, and currently underexplored, question.

Secondly, the potential for XAI methods to provide utility beyond simple post-hoc qualitative insights remains untested. A growing body of research has demonstrated that feature attributions are not static over time, but rather they can shift systematically between periods. Distinct patterns in feature attributions have been detected across market regimes and between periods of strong and weak model performance. It has been suggested that such patterns can act as indicators to practitioners of periods of reduced model reliability. However, no existing work has attempted to operationalise these insights into an automated real-time signal of prediction reliability. This represents the primary research gap this thesis attempts to address: whether systematic shifts in local XAI attribution patterns can be operationalised as a confidence signal to improve risk-adjusted trading performance.

1.2. Research Objectives

The primary objective of this thesis is to assess whether XAI feature attributions can be operationalised as real-time trading signals to enhance risk-adjusted trading performance. Additionally, it aims to compare more recently developed XAI algorithms to established methods, evaluating how they perform across a variety of model types within the specific domain of commodities financial time-series.

To achieve this, a diverse set of ML returns forecasting models are first built using a rich energy commodities feature set. A series of experiments then utilise these forecasting models to achieve the stated objectives. Recent ML architectures such as large language models and multi-agent systems are not considered because XAI methods are not yet well-established for such architectures, and data leakage is difficult to prevent due to these models being pre-trained on real-world data. Their inclusion would have shifted the focus from XAI analysis towards model engineering. Applicability of XAI to these models is identified as a direction for future work. However, TabPFN-3 – a recently released pre-trained transformer-based foundation model – is included in the model set. It is trained purely on synthetic data, alleviating concerns about data leakage, and is designed to operate cleanly on tabular data, allowing standard model-agnostic XAI methods to be easily applied. TabPFN’s inclusion provides an opportunity to test XAI method performance on newer model architectures rarely evaluated in the literature.

The experiments aim to compare the feature importance rankings produced by various XAI methods, and determine which methods perform best in this domain by evaluating their faithfulness and stability. In doing so, they also seek to provide valuable insight into which features are most important for forecasting energy price indices returns. The selection of XAI methods intentionally includes established methods (SHAP and LIME) and a recently developed time-series specific method (WinIT), so performance improvements from these recent developments can be benchmarked.

Furthermore, the experiments aim to identify patterns in local feature attributions across periods of high and low model performance, and to exploit these patterns as real-time

signals of prediction reliability – filtering which predictions a trading strategy acts on. By backtesting this strategy and comparing risk-adjusted performance to benchmarks, the ultimate aim is to assess whether XAI methods can be operationalised as real-time confidence signals.

1.3. Research Experiments

The thesis begins with an evaluation and comparison of the three selected XAI methods across the ML models, with the focus on global explanations. It then concentrates on local explanations, identifying patterns in feature attributions across periods of high and low model performance. Finally, it concludes with an out-of-sample backtest of trading strategies incorporating these identified patterns as real-time confidence signals. It is structured around three experiments:

Exp. 1 – Comparison of Explanations Across Methods and Models: This experiment analyses global feature importance across the XAI methods and ML models, evaluating the performance of the methods within the domain of energy price index returns on a diverse set of model architectures. SHAP, LIME, and WinIT are applied to generate global XAI explanations for each model on a 4-year test set, before identifying trends in what features the different models rely on, and discussing which features are most valuable for forecasting BCOMENTR returns. Cross-method and cross-model consistency is quantified using rank correlation. The XAI methods’ reliability is then evaluated with respect to these global explanations. Faithfulness is evaluated via faithfulness correlation to test whether XAI-attributed importance corresponds to actual predictive contribution. Stability is assessed using Gaussian-perturbed inputs, producing a quantitative robustness score per method-model pair. This experiment verifies which XAI methods produce trustworthy explanations within this specific domain. It also benchmarks recent XAI developments and applies XAI methods to more advanced ML models than is typical in the literature.

Exp. 2 – Local Attribution Patterns Across Prediction Outcomes: This experiment examines local attribution patterns across prediction outcomes. Local XAI explanations are computed for each out-of-sample prediction, including across the time dimension for the LSTM model. Attributions are then compared between incorrect and correct individual predictions. This experiment identifies whether systematic shifts in feature importance or temporal lag structure are detectable during model errors. Patterns are identified and tested for statistical significance. Of particular interest is whether LSTM models exhibit unrealistic temporal attribution profiles during incorrect predictions, and whether cross-method agreement degrades during loss-making periods. These attribution shift patterns form the basis of the confidence signal constructed in Experiment 3.

Exp. 3 – XAI-Derived Trading Signal: In this experiment, the findings of the second experiment are operationalised into a novel XAI-derived confidence-gating mechanism.

First, quantitative methods are designed for automatically recognising the unique error patterns in feature attributions identified the second experiment. Using this, the local attribution vector of each out-of-sample prediction is compared against the characteristic error-state profiles, informing the strategy of which predictions to ignore when making long/short decisions. Performance is backtested on an additional held-out test-set (January 2024 onwards) and benchmarked against an all-predictions strategy and a buy-and-hold baseline. Standard performance metrics including Sharpe ratio, Calmar ratio, and Max Drawdown are used. This potentially constitutes the first empirical application of XAI attribution patterns as a real-time confidence gate – a research direction inspired by recent literature.

1.4. Contributions

This research makes the following contributions to academia:

- 1) **Cross-method and cross-model evaluation of faithfulness and stability:** This thesis presents a quantitative evaluation of SHAP, LIME, and WinIT across three distinct model architectures, using standardised faithfulness and stability metrics. By including recently developed time-series-specific methods alongside established approaches (which have received the majority of attention within the discipline), this benchmark contributes empirical evidence on whether recent XAI developments offer meaningful improvements in this domain.
- 2) **Application and evaluation of XAI methods on energy commodities financial time-series:** Existing XAI benchmarks are predominantly conducted on general-purpose datasets. This thesis evaluates the behaviour of XAI methods in a less studied domain – energy commodities financial time-series – assessing how such methods handle the unique challenges of financial time-series, and this asset class in particular: low signal-to-noise ratio, non-stationarity, sequential structure.
- 3) **Application of XAI methods to TabPFN-3:** While previous literature predominantly applied XAI methods to standard ML models such as LSTMs and CNNs, this study applies them to TabPFN-3 – a pre-trained transformer-based in-context learning foundation model. Foundation models of this class are rarely evaluated in the XAI literature, and their inclusion provides insight into how explanation methods behave on frontier model architectures.
- 4) **Characterisation of local XAI attribution shifts under model error:** This thesis identifies and quantifies systematic patterns in local feature attributions between correct and incorrect predictions. These findings contribute to the empirical evidence that attribution patterns carry information about model reliability and are helpful for diagnosing causes of poor prediction performance.
- 5) **Novel XAI-derived confidence-gating mechanism for algorithmic trading:** This thesis introduces a method for converting local XAI attribution patterns into a real-time confidence signal. This operationalisation of XAI beyond simple post-hoc interpretation represents a novel contribution not previously tested in the literature.

Furthermore, this thesis makes the following contributions to industry:

- 1) **Feature importance rankings for energy commodity trading desks:** The global and local XAI analysis produces financially interpretable feature importance rankings for energy price index forecasting, identifying which macro, market, inventory, and cross-asset features can provide valuable trading signals.
- 2) **XAI method selection guidance for practitioners:** The faithfulness and stability analysis provides empirical guidance on which XAI methods are most reliable for financial time-series across different model architectures. This addresses a practical need of quant teams integrating XAI into model validation workflows under model risk management frameworks such as SR 11-7.
- 3) **Practical confidence-gating framework compatible with existing systematic strategies:** The XAI-derived confidence signal is designed to operate as an overlay on any trading strategy, requiring no model re-specification or retraining. This makes it directly deployable within existing trading systems, with the potential to improve risk-adjusted returns and reduce drawdown without altering the underlying alpha model.

1.5. Thesis Structure

The remainder of this thesis is structured as follows:

Chapter 2 – Background and Literature Review: This chapter outlines the key concepts and relevant literature related to the objectives of this thesis. It begins with an outline of the algorithmic trading field and how ML models have been applied in this domain. The XAI landscape is then presented, including prominent XAI methods and the standard evaluative criteria applied to them. It reviews the prior literature relevant to the research objectives, presenting the existing landscape and research gaps, and thereby giving the objectives academic grounding. Finally, the technical background of the ML architectures and XAI methods applied within this thesis are presented.

Chapter 3 – Data and Methodology: This chapter describes the data and methodology underpinning the empirical work. It presents the dataset, detailing data sources, target variable, feature engineering, stationarity transforms, and preprocessing. The train/test split and feature selection procedure are then outlined. The model validation procedure follows, covering the time-aware cross-validation approach used for hyperparameter optimisation. Finally, the implementation of each ML model and XAI method is specified, including software packages and hyperparameter configurations.

Chapter 4 - Exp. 1: Comparison of Explanations Across Methods and Models: This chapter presents the first experiment. It begins by precisely defining the faithfulness and stability criteria used to evaluate XAI explanations, before applying SHAP, LIME, and WinIT across all three models to generate global feature importance rankings. The chapter concludes by evaluating each method-model pair against these criteria,

identifying which methods produce trustworthy explanations in the energy commodities returns domain.

Chapter 5 - Exp. 2: Local Attribution Patterns Across Prediction Outcomes: This chapter presents the methodology and results of the second experiment, beginning with the generation of local XAI explanations for each out-of-sample prediction. Attributions are then compared between incorrect and correct individual predictions, and patterns are identified and tested for significance.

Chapter 6 - Exp. 3: XAI-Derived Trading Signals: This chapter presents the third experiment. First, quantitative methods are developed for automatically recognising the unique error patterns in feature attributions identified in the second experiment. These are then implemented as a confidence signal in a trading strategy. The performance of this XAI-derived model is backtested on a held-out test-set (January 2024 onwards) and benchmarked against an all-predictions strategy and a buy-and-hold baseline.

Chapter 7 - Conclusion and Future Work: The final chapter summarises the key findings of this study and provides an overall conclusion with reference to the key research objectives set out earlier. It also discusses the key contributions derived from the experiments and their results. The thesis ends with a discussion of the limitations of this study and proposals for future research directions in this area.

Chapter 2

Background and Literature Review

This chapter provides background information on the key topics and concepts relevant to this thesis, referencing the existing literature on these topics. The first sub-chapter reviews research focussed on similar applications of XAI methods (within financial time-series in general, and more specifically for forecasting returns and applied to energy commodities). It concludes with the areas that are ripe for further investigation, presenting the key gaps in literature that this thesis attempts to fill. The second and third sub-chapters provide background on the ML models and XAI methods used within this thesis respectively, including their mathematical and theoretical foundations, and their documented strengths and limitations.

2.1. Literature Review

2.1.1. Machine learning for energy commodities trading

The energy commodities trading field has followed a similar trajectory to the wider algorithmic trading and financial forecasting discipline. Both academic and industrial practices have employed a wide variety of techniques, ranging from econometric models and classical statistical time-series methodologies to more recent machine learning approaches.

Earlier research focused on applying traditional statistical and econometric models. For example, Wen, et al. (2019) used a VAR model with time-varying parameters for forecasting oil using financial factors (e.g. USD value, S&P 500 index), while Kilian & Baumeister (2015) forecasted oil prices by combining six econometric models that incorporated factors ranging from oil futures to gasoline and heating oil spread. Analysing the key factors influencing crude oil markets was also a common route towards improving forecasting performance using traditional econometric techniques. Supply and demand side fundamental factors have been identified as the most influential, including oil supply shocks and inventory levels, and global demand shocks and oil consumption trends – including in the wider industrial commodities market (Kilian, 2009) (Kilian & Murphy, 2013). Since 2008, financial factors (e.g. speculation, investor sentiment, global liquidity) have also played a significant role, particularly with regard to short-term volatility (Liu, et al., 2018).

More recently, machine learning and deep learning techniques for financial forecasting have proliferated, as advances in these areas have spread from the computer science domain into the quantitative finance field. Several influential papers have driven this trend. Gu, et al. (2020) demonstrated large predictive and economic gains from using machine learning forecasting methods such as trees and neural networks compared to

traditional regression-based strategies. This highlighted that non-linear interaction effects of input features can be harnessed to improve prediction performance. Meanwhile, Fischer & Krauss (2018) found LSTM network models outperform traditional logistic regression classifiers for forecasting daily directional movements, as well as memory-free ML methods (a random forest and a deep neural network). These influential papers also established more rigorous frameworks for applying and evaluating ML techniques in financial forecasting, such as time-aware cross validation and trading performance evaluation via backtesting.

These seminal papers focussed on equities markets, but the drive towards ML and DL techniques has quickly spread across asset classes, including into commodities and crude oil specifically. The breadth of research on commodities is significantly smaller (Sezer, et al., 2020), but several studies – spanning different model comparisons, time horizons, and feature sets – provide strong evidence that ML techniques offer stronger predictive performance than traditional statistical techniques. Herrera, et al. (2019) compared an artificial neural network and a random forest model to a hybrid econometric model and a no-change baseline for energy forecasting at a monthly frequency (including for oil). By comparing RMSE and MAPE, they found the ML models significantly outperformed the econometric and baseline models, with random forests performing the best. Similarly, Göncü, et al. (2024) compared six ML models for forecasting daily crude oil return direction. Again, they found that the ML models generally outperformed the logistic regression baseline, with the CNN models consistently displaying the best directional accuracy. The ANN model performed comparably well at shorter forecast horizons, while the XGBoost model did so at longer forecast horizons. Foroutan & Lahmiri (2024) recently compared 16 ML and DL methods for forecasting daily crude oil, gold, and silver prices. They found that a temporal convolutional network and a LightGBM model performed best on crude oil, with similar results for the precious metals. Other studies have found success incorporating a richer set of features with ML models, with Wang & Chen (2025) integrating sustainability and external risk variables for crude oil forecasting. In that study, the XGBoost model was the standout for prediction performance. As illustrated by these papers, ML models have consistently outperformed traditional statistical models, validating their increased adoption. For studies that forecast crude oil returns (rather than prices), directional accuracy is the most common evaluation metric, and performance of 54% to 60% is typical of most models (Göncü, et al., 2024) (Wang & Chen, 2025).

Generally, the commodity trading literature has trended in the same direction as the wider algorithmic trading discipline: towards increasingly complex ML techniques. Earlier work focussed on traditional ML methods like support vector machines, random forests and GBMs, before the field progressed towards DL methods like DNNs, RNNs and LSTMs, and CNNs (Bhuiyan, et al., 2025), and very recently introduced transformers and reinforcement learning (RL) agents (Dakalbab, et al., 2024). ML models have consistently been shown to outperform classical statistical models, while the evidence

on DL models specifically (e.g. DNNs, LSTMs, CNNs, and RL models) compared to GBMs is more mixed, as illustrated by the previously discussed studies. However, this movement towards increasingly complex models has produced a new problem: interpretability. While these models have superior performance, it is more difficult to understand how they are making decisions or what patterns they have extracted from the data. This is a major barrier to trust for commodity traders and financial regulators (Bhuiyan, et al., 2025) (Yeo, et al., 2025), and has led to the development of explainable AI (XAI).

2.1.2. Explainable AI – methods and evolution

The increasing complexity of ML and DL models has been accompanied by increasing opacity in their internal decision-making processes. This has impeded practitioners from using these models – despite their often superior performance – due to difficulties understanding, trusting, and auditing them. This friction gave rise to Explainable AI (XAI), broadly defined as a suite of ML techniques that enable stakeholders to understand, appropriately trust, and effectively manage ML models (Adadi & Berrada, 2018). The aim is to increase explainability and understanding without sacrificing the superior predictive performance.

Early solutions to this problem focussed on models that are inherently interpretable – commonly described as white-box or transparent models – including linear and logistic regression, decision trees, k-nearest neighbours, rule-based learners, and Bayesian models. As predictive tasks demanded increasingly complex black-box architectures, such as tree ensembles and deep neural networks, this intrinsic route to interpretability became infeasible, and attention shifted toward post-hoc explainability – techniques applied after training to explain a black-box model's behaviour without altering its internal structure.

Two contributions are widely credited with pushing post-hoc XAI into the mainstream, and have become the most widely adopted XAI methods across ML domains: Local Interpretable Model-Agnostic Explanations (LIME) (Ribeiro, et al., 2016), which approximates a black-box model locally around a given prediction using an interpretable surrogate model, and SHapley Additive exPlanations (SHAP) (Lundberg & Lee, 2017), which draws on cooperative game theory to attribute a prediction's output to its input features. A key strength of both methods, and a major reason for their widespread adoption, is their model-agnostic nature – because model-agnostic techniques are not tied to a particular type of model, the same method can be applied across model architectures. This versatility is also what makes the cross-model comparison within this thesis possible.

Since these two landmark papers, a wide range of XAI methods have been developed with distinct methodologies, strengths, and weaknesses. They can be broadly categorised according to three criteria (Barredo Arrieta, et al., 2020) (Adadi & Berrada, 2018) (Dwivedi, et al., 2023) (Bodria, et al., 2023):

- **Model-agnostic or model-dependent/model-specific:** Model agnostic methods can provide explanations for any ML model, while model-dependent/model-specific methods are designed to operate on a single class of models, dealing with the inner workings of a model to interpret its results. Both LIME and SHAP are model-agnostic, while Integrated Gradients (IG) is an example of a model-dependent method since it can only operate on neural networks.
- **Local or global explainability:** Local explanations analyse individual predictions, clarifying why those predictions were made and focussing on a subregion around that data point. Meanwhile, global explanations describe a model's overall behaviour, not pertaining to any specific data point or prediction. LIME provides local explanations while SHAP provides both local and global explanations.
- **Methodological class:** post-hoc XAI methods can generally be grouped into several classes based on their theoretical approach to explainability. The most common classes include:
 - > *Surrogate models* – methods such as LIME that fit a simpler inherently interpretable model to the black-box model's predictions to approximate its decision-making process.
 - > *Occlusion methods* – methods that mask individual features, replacing them with baseline values, and then measuring the change in output. Feature Occlusion is an example.
 - > *Perturbation/permutation based* – methods such as SHAP that mask subsets of input features to remove any meaningful signal, then measure the change in model output across many permutations.
 - > *Gradient-based* – methods that analyse the weights within a neural network to assess how different features influence the final output and how they interact. IG is an example of this.
 - > *Counterfactuals* – methods that present the minimum change in the input features necessary to change the model's prediction. One example is Diverse Counterfactual Explanations (DiCE).

Time-series forecasting presents unique challenges for many XAI methods. Standard LIME/SHAP perturbation schemes assume features can be independently masked/permuted without disturbing the underlying data-generating process (Aas, et al., 2021). However, this assumption does not hold in time-series data due to temporal autocorrelation (time steps are not independent) – masking timestep t while leaving $t-1$ and $t+1$ untouched produces an unrealistic, out-of-distribution sample. This concern has led to the recent development of specialised time-series-specific XAI methods within each methodological class.

One such method is implemented within this thesis: WinIT (Leung, et al., 2023). This is an occlusion-based method which measures the marginal contribution of each time lag for each feature, using a sliding window approach to account for the autocorrelation present in between lags. The technical details of all methods used within this thesis are

presented in Section 2.3, while their specific implementations are detailed in Section 3.4.2.

2.1.3. Evaluation of XAI methods

As the need for post-hoc XAI for explaining complex models in high-stakes situations has grown, and the number of XAI methods available has expanded, there has been increasing demand for methods to evaluate XAI methods. Different methods not only offer different types of explanations, but often disagree in their explanations (Krishna, et al., 2025) and also perform differently across datasets and model architectures (Agarwal, et al., 2022) (Mercier, et al., 2022). Therefore, evaluation is necessary to ensure explanations are of good quality and appropriate for a user’s objectives, and to aid users in selecting the best XAI methods for a given situation.

A broad range of XAI evaluation criteria and metrics have been developed. There have been many competing frameworks and taxonomies proposed for organising these (Zhou, et al., 2021) (Agarwal, et al., 2022) (Nauta, et al., 2023) (Bodria, et al., 2023) (Kadir, et al., 2023), and there is no clear consensus on which evaluative criteria or metric categories are most relevant. However, two that are most common are faithfulness (also referred to as fidelity or correctness) and robustness (also referred to as stability). Faithfulness measures the extent to which explanations match the true behaviour of the model, assessing whether more important features have greater effects on model outputs. Robustness measures whether explanations are stable under small random perturbations to the input data (Alvarez-Melis & Jaakkola, 2018). These are the two evaluation criteria used within this thesis. Their formal definitions and implementations are presented in Section 4.2.

A wide range of other evaluation criteria have been proposed by various papers, such as fairness, complexity, compactness/parsimony, runtime, and many more (Zhou, et al., 2021) (Bodria, et al., 2023) (Nauta, et al., 2023). However, these are less common and less universal. Human-centred criteria relating to presentation and human experience of explanations have also been proposed, but measuring these would require designing experiments involving humans. Therefore, these are not considered within this thesis.

There is evidence that SHAP and LIME perform well in terms of fidelity/faithfulness, but LIME has been shown to be unstable while SHAP has better stability (Agarwal, et al., 2022) (Mercier, et al., 2022). However, both methods perform worse on time-series datasets for the reasons explained in Section 2.1.2. The more recently developed XAI methods like WinIT have received less scrutiny, which is a gap this thesis attempts to fill.

There is also evidence that performance varies significantly depending on datasets and domains (Agarwal, et al., 2022) (Mercier, et al., 2022). Therefore, this thesis tests these XAI methods on an energy commodities dataset specifically – a less common domain for their evaluation – to contribute to the diversity of the body of research on their performance.

2.1.4. XAI for energy commodities forecasting

XAI has been readily adopted within ML financial forecasting literature in the last few years as researchers attempt to integrate transparency into their models. However, most literature has focussed on credit management or fraud detection, and even for those concerning price/return forecasting, the vast majority focus on equities (Černevičienė & Kabašinskas, 2024). Even so, some papers have applied XAI in the context of energy commodities and crude oil specifically. Guliyev & Mustafayev (2022) apply SHAP to their XGBoost monthly return forecasting model to measure global feature importance. Among their set of macroeconomic and financial features, the 5 most important were: refiner acquisition cost, futures contract volume, USD index, unemployment rate, oil/gas industrial production index. Jabeur, et al. (2021) apply SHAP to their highest performing COVID-19 oil crash prediction models at each forecasting horizon (a random forest model at 7, 14, 21 days-ahead, and an XGBoost model at 1-month ahead). Their results show that the importance of some features shift significantly depending on the forecasting horizon, although this finding is slightly conflated with the change of model at 1-month ahead. Gao, et al. (2022) applied SHAP to their LightGBM daily price forecasting model. Their global analysis found the S&P 500 and Bitcoin prices to be most influential and validated classical economic theory regarding oil inventories negatively correlating with oil prices. Meanwhile their SHAP dependence plots proved most interesting by illustrating the non-linear relationships between features like silver and crude oil prices. Most recently, Wang & Chen (2025) applied SHAP to their XGBoost returns forecasting model, finding rate of change (a momentum indicator) and VIX to be the most important features. Unlike the earlier literature which only applied SHAP to their highest performing models, they also compared explanations between models. All the ML and DL models displayed similar feature rankings, while OLS differed significantly – likely due to its inability to extract non-linear relationships between features.

While most literature in this domain has focussed exclusively on global feature importance – identifying the most globally influential features and assessing whether these align with financial/economic intuition – a few have identified more interesting time-varying trends. Wang & Chen (2025) computed local explanations across their full sample, and found a collapse in model accuracy post-2020 corresponded with a sharp drop in importance for some of the previously most influential features. While feature rankings were stable until 2020, some macroeconomic and financial indicators, such as VIX, USD exchange rates, and the Chicago Fed’s National Activity Index, dropped significantly in importance during and after COVID-19, while technical indicators and GDP increased in prominence. It is worth noting that the authors suggest this shift indicates a change in market structure, implicitly assuming the model and explanations are accurate representations of the market during this period, and giving no consideration to the possibility of it being an indicator of model failure. In contrast, Rettinger, et al. (2025) also identified feature importance shifts over time and explicitly frame these as indicators of model reliability. They constructed commodity hedging

models (rather than returns forecasting models), with crude oil and natural gas as two of their four target commodities, and compared SHAP values between two out-of-sample periods: a stable 2018-2020 period, and an unstable 2021-2023 period where model performance dropped. Explanations for crude oil were not explicitly presented, but natural gas explanations were shown and noted to be structurally similar to those of crude oil. During the unstable period, SHAP magnitudes are found to collapse towards 0 and feature rankings become unstable. Furthermore, time-step attributions invert compared to the stable period. However, while these papers have raised the possibility of XAI explanations as indicators of prediction reliability, they have never operationalised that idea as a real-time confidence signal for a trading strategy.

The idea of operationalising XAI explanations within a trading strategy, beyond simple post-hoc analysis, is relatively novel, with only two existing papers experimenting with similar ideas. Çelik, et al. (2023) and Moustati & Gherabi (2026) both treat XAI outputs as confidence-filtering signals rather than diagnostic tools. Çelik et al. extract per-prediction class probabilities from LIME's local surrogate model of a Random Forest's stock price direction predictions, and abstain from trading unless that probability exceeds a chosen threshold. This produced an increased accuracy, climbing from ~80% to mid-90% as coverage shrinks to a fraction of trading days. Moustati & Gherabi push this further, converting TimeSHAP and TIME explanations on an LSTM into continuous confidence scores that influence stop-loss/take-profit bands and position sizes within an RL trading agent. They report a large improvement in risk-adjusted returns (Sharpe of 2.11) over rule-based and buy-and-hold benchmarks on TSLA. However, they offer little justification for their choice of confidence signal, and only test on a single stock (TSLA) over a single year. This thesis aims to develop this area of research by providing rigorous justification for the confidence score filtering formulation through in-depth analysis of feature attribution shifts.

A major weakness of most of the reviewed papers is that they implicitly assume the reliability of their explanations without attempting to assess explanation quality. Furthermore, most only use SHAP without considering any alternative XAI methods, a trend reflected in the wider financial forecasting literature where SHAP is dominant (Černevičienė & Kabašinskas, 2024). This is concerning due to SHAP's previously discussed limitations for time-series applications. Rettinger, et al. (2025) is the only exception within the energy commodities literature, comparing SHAP and LIME and assessing their explanation quality in terms of faithfulness and sanity. They then proceeded with SHAP for the core analysis because it outperformed LIME. This thesis follows a similar methodology, enhancing the methodological rigour of earlier work by applying well-defined quantitative XAI evaluative metrics. It goes further than Rettinger, et al.'s work by also incorporating a recently developed time-series specific method for comparison, and comparing across model architectures, including a tabular foundation model not previously tested in the existing literature.

2.2. Forecasting Models

2.2.1. XGBoost

XGBoost (Extreme Gradient Boosting) is a regularised, scalable implementation of gradient boosting, introduced by Chen & Guestrin (2016) as an extension of Friedman's (2001) gradient boosting machine. It is one of the dominant algorithms for structured, tabular prediction tasks, and is included in this thesis as a non-sequential, tree-based benchmark. Gradient-boosted trees, and XGBoost specifically, have an established track record in commodity price forecasting, and for financial forecasting in general. They are among the most popular models for such tasks, further justifying their inclusion in this study.

MATHEMATICAL FORMULATION

An XGBoost model comprises an ensemble of decision trees, and generates a prediction as the sum of outputs from K trees $f_k(x)$:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i).$$

During training, trees are added one at a time. The current ensemble's prediction $\hat{y}_i^{(t-1)}$ is held fixed, and a new tree f_t is fitted to predict the ensemble's remaining error, so that $\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(x_i)$. The tree at each round is chosen to minimise a regularised objective function:

$$\mathcal{L}(t) = \sum_{i=1}^n L(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t),$$

where $L(y_i, \hat{y}_i^{(t-1)} + f_t(x_i))$ is the unregularised loss function, $\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2$, T is the number of leaves in tree f_t , and w is the vector of leaf weights. Since this thesis models forecasting as a regression task, the loss function used is the mean squared error.

The regularisation term Ω – absent from Friedman's original formulation – penalises both the number of leaves and the magnitude of their weights, and is XGBoost's central methodological departure from standard gradient boosting: it discourages overly complex trees directly within the objective, rather than relying solely on shrinkage or early stopping for generalisation.

XGBoost approximates the loss function via a second-order Taylor expansion of the loss around the current prediction. Dropping terms constant with respect to f_t gives:

$$\mathcal{L}(t) = \sum_{i=1}^n \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t(x_i)^2 \right] + \Omega(f_t),$$

where g_i and h_i are the first- and second-order gradients of the loss with respect to $\hat{y}_i^{(t-1)}$.

Whereas Friedman's original algorithm used only g_i – fitting each tree by least-squares regression to the negative gradient, then determining a step size separately – XGBoost also uses h_i to characterise the local curvature of the loss around the current prediction. This is the function-space analogue of Newton's method: rather than taking a step of some externally chosen size in the direction of steepest descent, the curvature term allows the exact minimiser of the local second-order approximation to be solved for directly, for each leaf, in closed form.

To compute the optimal leaf weights to minimise the loss function, note that once the tree's structure (its splits) is fixed, every observation falling in leaf j receives the same output: $f_t(x_i) = w_j$ for all $i \in I_j$, where I_j is the set of observations in leaf j . This lets the sum over observations be restated as a sum over leaves, with the per-leaf regularisation substituted in directly:

$$\begin{aligned}\mathcal{L}(t) &= \sum_{j=1}^T \left[(\sum_{i \in I_j} g_i) w_j + \frac{1}{2} (\sum_{i \in I_j} h_i) w_j^2 \right] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2, \\ &= \sum_{j=1}^T [G_j w_j + \frac{1}{2} (H_j + \lambda) w_j^2] + \gamma T,\end{aligned}$$

for $G_j = \sum_{i \in I_j} g_i$ and $H_j = \sum_{i \in I_j} h_i$.

Differentiating with respect to w_j produces the optimal leaf weight w_j^* :

$$w_j^* = -\frac{G_j}{(H_j + \lambda)},$$

which can be substituted back into the loss function to give leaf j 's best-achievable contribution to the loss, $-\frac{G_j^2}{2(H_j + \lambda)}$, which forms the basis of a structure score used to evaluate candidate splits during tree construction. For a split partitioning instance set I into left and right children I_l and I_r , the gain is:

$$Gain = \frac{1}{2} \left[\frac{G_l^2}{(H_l + \lambda)} + \frac{G_r^2}{(H_r + \lambda)} - \frac{(G_l + G_r)^2}{(H_l + H_r + \lambda)} \right] - \gamma.$$

Candidate splits are tested over every feature and for every possible threshold value for that feature, with the gain computed in each case. The candidate split with the greatest gain is then selected. This is applied recursively to each child node until a maximum depth is reached, no split has positive gain, or another stopping criterion is met. The resulting tree is added to the ensemble, and the process repeated until K trees are built.

2.2.2. LSTM

Long Short-Term Memory (LSTM) networks are a recurrent neural network (RNN) architecture introduced by Hochreiter & Schmidhuber (1997), with the now-standard forget gate added by Gers, et al. (2000). Unlike XGBoost and TabPFN, which treat each week's feature vector as an independent, exchangeable observation, LSTM processes the input as an ordered sequence, maintaining an internal state that is updated at every time

step. It is included in this thesis as the temporally-aware benchmark, and enables the application of time-series-specific XAI methods such as WinIT.

MATHEMATICAL FORMULATION

Figure 2.1 displays the structure of a single cell in an LSTM. The equations below describe the processing happening during each cell state.

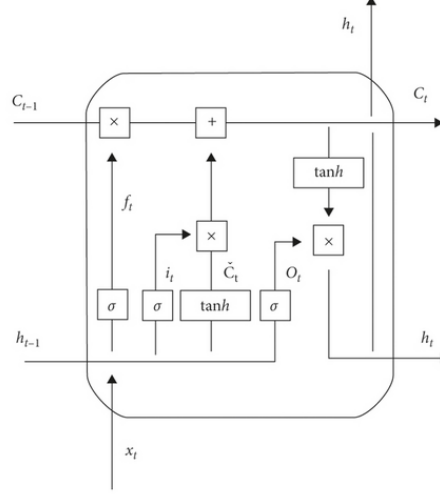


Figure 2.1: Structure of an LSTM cell (Jung & Choi, 2021)

This cell will process each lag provided in the input data sequence using the information retained from processing previous lags. x_t is the new input data at lag t . h_{t-1} and c_{t-1} are respectively the hidden state and memory state passed forward from the processing of the previous time lags. The hidden state is the actual output of that previous process while the memory state is additional information that is passed forward. They will be initialised to $h_0=c_0=0$. x_t and h_{t-1} together form the current information $\tilde{c}_t$ flowing into the cell:

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c),$$

The gates f_t , i_t , and o_t are the forget gate, input gate, and output gate respectively. They are all also dependent on x_t and h_{t-1} , and are all composed of sigmoid functions $\sigma(\bullet)$, as described by the following equations:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f),$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i),$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o).$$

The forget gate determines the weighting of the past memory state that gets kept in the new present memory state. The input gate determines the weighting of the current information flow that gets incorporated into the memory cell state. Combined, these produce the current memory state c_t :

$$c_t = (f_t \times c_{t-1}) + (i_t \times \tilde{c}_t).$$

This memory state gets passed on to the next cell state. It also interacts with the output gate to produce the new hidden state (which is the output for this lag in the sequence):

$$h_t = o_t \times \tanh(c_t).$$

A stacked LSTM comprises multiple LSTM layers. Each layer consists of an LSTM cell which iterates through each lag in the process described above. The outputs $h_1, h_2, \dots$ from one of these layers are the inputs $x_1, x_2, \dots$ into the next layer. A stacked structure allows the network to learn complex temporal relationships over long time horizons.

Dropout layers can be included between each LSTM layer as a form of regularisation to reduce the risk of overfitting. During each training iteration, a dropout layer with a dropout probability p will randomly select p proportion of the nodes in the layer to remove. The outputs of the remaining neurons are scaled up by a factor of $\frac{1}{1-p}$.

Training is conducted by backpropagation through time: the network is unrolled across the input sequence, and gradients are computed via the chain rule applied to this unrolled graph, using a first-order stochastic optimiser (Adam). While LSTM substantially mitigates vanishing gradients, exploding gradients remain possible during training, and gradient clipping is commonly applied as a further safeguard.

2.2.3. TabPFN

TabPFN (Tabular Prior-data Fitted Network) is a transformer-based foundation model for tabular prediction, built on the Prior-Data Fitted Network (PFN) framework of Müller, et al. (2022), and introduced for tabular classification by Hollmann, et al. (2023) (2025). Unlike XGBoost and LSTM, TabPFN is not fitted to this thesis's data at all, but rather is pre-trained on millions of synthetic datasets. As an in-context learning foundation model, it produces predictions for any new dataset via a forward pass alone. It is included as the third, structurally distinct paradigm in the model stack, representing the frontier of foundation model approaches for tabular prediction.

This thesis uses the regression variant of TabPFN-3 – the most recent release – which improved on TabPFN v2 with a larger context capacity and faster inference. It was released in May 2026, making it – by a wide margin – the newest model in the stack. TabPFN-3 is documented only in a technical report (Grinsztajn, 2026) rather than a peer-reviewed publication at the time of writing, but the previous version is documented in an influential peer-reviewed paper (Hollmann, et al., 2025). TabPFN's inclusion reflects a deliberate emphasis on evaluating XAI methods on a frontier state-of-the-art architecture.

An important property of TabPFN is its pre-training on purely synthetic datasets, alleviating concerns about data leakage that other foundation models suffer from and that would make true out-of-sample evaluation impossible.

MATHEMATICAL FORMULATION

TabPFN models are transformer-based models, utilising the same attention mechanisms commonplace in the latest state-of-the-art deep learning models and large language models. This allows each element of a set to weight and influence information from every other element, rather than processing them in fixed positional order. Attention projects each element in an observation into query, key, and value vectors (Q, K, and V respectively) by multiplying by weight vectors W^Q , W^K and W^V respectively. The output for each element is a weighted sum of every element's value vectors, where the weights (attentions) come from comparing the original element's query against each element's key:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V.$$

The softmax function transforms the comparisons into weights that sum to 1. Dividing by $\sqrt{d_k}$ (the key dimension) keeps the dot products from growing too large as dimensionality increases.

TabPFN's two-way attention applies this twice per cell: once across features within a row, and once across observations within a feature (although only one way for train data to test data – there is no attention from test data to train data to ensure no data leakage).

The model is trained on millions of synthetic datasets built from structural causal models to reflect the characteristics of real-world data. Each dataset is generated by:

- (1) Randomly sampling key hyperparameters (e.g. number of features, causal graph complexity).
- (2) Setting initialising data for the initial nodes of the graph (using uniform or normal noise).
- (3) Propagating that data through the graph.
- (4) Randomly sampling which nodes will be features and which will be the target (leaving the rest ignored).
- (5) Post-processing, quantization, and warping of the selected nodes' data (e.g. removing values to replicate missing data).

Millions of such synthetic datasets are used, and this pretraining happens once, offline, entirely independently of any real-world datasets.

2.3. XAI Methods

2.3.1. LIME

Local Interpretable Model-agnostic Explanations (LIME) was introduced by Ribeiro, et al. (2016) as a model-agnostic post-hoc method for explaining individual predictions of any black-box classifier or regressor by approximating its behaviour locally with an

interpretable surrogate model. It produces exclusively local explanations with no native mechanism for aggregating these into a global account of model behaviour.

MATHEMATICAL FORMULATION

For a black-box model f and an instance x to be explained, LIME attempts to find the inherently interpretable model g that solves:

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}} [L(f, g, \pi_x) + \Omega(g)],$$

where G is a class of inherently interpretable models, L is a locally-weighted fidelity loss function, π_x is a measure of proximity of a datapoint to x , and $\Omega(g)$ is a penalty function to reduce complexity.

The standard choices for these – which are used in the LIME python module utilised within this thesis – are for G to be the class of linear models, L to be a weighted squared loss function $L(f, g, \pi_x) = \sum_{z, z' \in Z} [\pi_x(f(z) - g(z'))^2]$, π_x to be an exponential function such that $\pi_x(z) = e^{\left(-\frac{D(x, z)^2}{\sigma^2}\right)}$ for some distance function D , and $\Omega(g)$ to be the number of non-zero coefficients.

The solution to the above function is approximated via random sampling. The process is as follows:

- (1) A sample of perturbed datapoints is generated in the neighbourhood of x . This is done by sampling each feature independently from a normal distribution defined by that feature's mean and standard deviation in the training data.
- (2) Each perturbed datapoint is passed through f to obtain a prediction.
- (3) Each datapoint is weighted by π_x according to its distance from x .
- (4) A weighted linear model is fit to the (perturbed datapoint, prediction) pairs, with a penalty for complexity.

The resulting weighted linear model is g , and its coefficients are the feature importance values.

LIMITATIONS

LIME rests on the assumption that, within a sufficiently small neighbourhood, f 's decision surface is approximately linear. This assumption does not necessarily hold, and the quality of the explanation is sensitive to a neighbourhood (kernel bandwidth) that has no principled, data-driven method for selection.

Also, because explanations are estimated from randomly sampled perturbations rather than computed exactly, they carry sampling variance – repeated explanations of the same instance are not guaranteed to be identical unless randomness is explicitly controlled.

Most relevant to this thesis, LIME's default perturbation scheme samples each feature independently from its marginal distribution, implicitly assuming feature

independence. This is a poor assumption for time-series datasets, where adjacent lags are typically autocorrelated and independently perturbing each feature can generate samples that are unrealistic.

2.3.2. SHAP

SHAP (SHapley Additive exPlanations) is a model-agnostic, post-hoc attribution method based on the Shapley value (Lundberg & Lee, 2017) – a solution concept from cooperative game theory for fairly dividing a coalition's total payoff among its players. SHAP reframes a prediction as a cooperative game in which the input features are ‘players’ cooperating to produce the model's output, and asks how much of that output should be credited to each feature, with the payout distributed according to each feature's average marginal contribution across all possible feature coalitions. It produces both local explanations and, via aggregation, global explanations, due to its axiomatic additive decomposition.

For a model f and feature set F , the Shapley value for feature i is defined as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)],$$

where f_S is the expected model output when only the features in coalition S are known and the remaining features are marginalised out. The formula averages feature i 's marginal contribution over every possible order in which features could enter the coalition, which is what makes the resulting attribution independent of any single arbitrary feature ordering.

MATHEMATICAL FORMULATION

SHAP formalises the output as an additive feature attribution model:

$$g(z') = \phi_0 + \sum_{i=1}^M [\phi_i z'_i],$$

where g is a local linear approximation of f , ϕ_0 is the base value (expected output over the background distribution), M is the number of input features, and $z' \in \{0,1\}^M$ indicates feature presence.

This is the same additive-attribution structure LIME satisfies, but SHAP's coefficients are uniquely pinned down by three axioms: (1) local accuracy ($f(x) = \phi_0 + \sum \phi_i$), (2) missingness (absent features get zero attribution), and (3) consistency (a feature's attribution cannot decrease if its marginal contribution weakly increases under a model change). This axiomatic grounding – rather than heuristic local fitting – is SHAP's principal theoretical advantage over LIME.

SHAP produces one attribution vector per observation. Aggregating ϕ_i across the observations using the mean absolute SHAP value produces global importance values.

Exact computation of SHAP values would require evaluating f over all $2^{|\mathbf{F}|}$ subsets, which is impractical at realistic feature counts. KernelSHAP instead estimates ϕ_i via weighted linear regression: coalitions z' are sampled, f is evaluated with hidden features replaced by draws from a background dataset, and a linear model is fit using the same estimation framework as LIME but with $\Omega(g) = 0$ and the SHAP kernel weights:

$$\pi_x(z') = \frac{M - 1}{\binom{M}{|z'|} |z'| (M - |z'|)}.$$

This upweights coalitions with very few or very many active features, since these are most informative about individual marginal contributions. The fitted regression coefficients are the estimated Shapley values.

LIMITATIONS

Like LIME, KernelSHAP's perturbation step independently samples each feature, implicitly assuming feature independence. This can produce implausible, out-of-distribution combinations when features are correlated.

Additionally, the sampled coalition space introduces estimation variance. Repeated runs on an identical input can yield slightly different attributions purely from sampling noise.

Furthermore, attributions are sensitive to the choice of background dataset, which is fixed here via a k-means summary of the training set rather than the full training distribution.

2.3.3. WinIT

WinIT (Windowed Feature Importance in Time), introduced by Leung, et al. (2023), is a feature-removal-based saliency method developed specifically to address two challenges left unresolved by applying LIME or SHAP directly to time-series data: observations of the same feature at nearby time steps are temporally dependent, so perturbing them independently produces implausible, out-of-distribution samples. WinIT is applicable only to sequential models – only the LSTM in this thesis.

MATHEMATICAL FORMULATION

For a feature subset S at time a , WinIT produces an importance score $I(S, a, b)$ quantifying S 's specific contribution to the prediction made at a later time b , and an aggregate score $\mathcal{I}(S, a)$ summarising S 's importance across a forward-looking window of predictions. Explanations are local: a full (feature $\times$ time-step) importance matrix is produced per observation, with no native mechanism for global aggregation, as with LIME.

For a model f_θ producing a predictive distribution $p(y_b | X_{\{1:b\}})$ at each time step b , the importance of feature set S to that prediction, computed over the interval $[a, b]$, is defined by comparing:

$$P = p(y_b | X_{\{1:b\}}), \text{ and } Q = p(y_b | X_{\{a:b\}}^{\sim S}),$$

where $X_{\{a:b\}}^{\sim S}$ denotes the input series with feature set S removed across time steps a through b .

Leung et al. consider three divergence measures between P and Q – prediction difference, KL divergence, and Jensen-Shannon divergence – and adopt prediction difference based on empirical comparison:

$$i(S)_a^b = \|P - Q\|_1.$$

Because subsequent observations x over time are temporally dependent, the importance of S over $[a,b]$ alone conflates lag a 's contribution with every later lag still contained in that interval. WinIT isolates a 's unique contribution by taking a difference between two overlapping windows:

$$I(S, a, b) = i(S)_a^b - i(S)_{a+1}^b \text{ for } a < b < a + N,$$

$$I(S, a, a) = i(S)_a^a.$$

Removing S from $[a+1,b]$ captures the combined importance of every observation after a , so subtracting this from the importance of removing S from $[a,b]$ isolates what observation a specifically contributes, beyond what more recent lags already explain (the marginal contribution).

Chapter 3

Data and Methodology

This chapter describes the datasets used. It then describes and justifies all manipulation done to the data, including stationarity analysis and transformations, test-train split, feature selection, and scaling.

3.1. Data

This study uses a time series dataset consisting of a full range of standard energy commodities forecasting features. These include general macroeconomic and financial market features, as well as crude oil specific features identified using the commodity playbook function on Bloomberg (CPLY). All data is sourced from Bloomberg.

One additional feature was generated to incorporate high and low information on WTI, in addition to the weekly open and close implicitly included by the time-series. This feature was CL1 H/L, computed by dividing the weekly high by the weekly low of the WTI price. The full set of features is listed in Appendix B alongside their summary statistics.

All time-series are converted to a weekly frequency, with monthly frequency data forward filled to avoid lookahead bias.

The target variable is the Bloomberg Energy Subindex Total Return (BCOMENTR). This index tracks futures contracts on crude oil, heating oil, unleaded gasoline and natural gas. It is highly correlated with crude oil futures, but alleviates difficulties associated with using raw crude oil futures which have to be rolled over each month to maintain consistency. The price series is converted to a weekly returns series to ensure stationarity. The resulting returns series is checked to confirm it is balanced between positive and negative returns. This avoids the class-imbalance problem, in which models learn to predict predominantly in a single direction, achieving high performance without learning any real patterns in the data.

The initial data analysis and feature engineering is conducted on the full history of each feature, going back as far as 01/01/1985 where possible. Once feature selection is completed according to the process described in Section 3.2.2, the final time period of 23/04/2004 – 31/12/2025 is selected as the maximum range without any missing data for the selected features. This results in 1133 total weekly observations.

The full dataset is split into a training set and two test sets. The training data is used to train all three forecasting models. The first test set is used for testing model performance

and conducting the XAI analysis in Experiments 1 and 2. The second test set is used in Experiment 3 to evaluate the performance of the final trading strategy. Due to the temporal autocorrelation present in time-series data, random splitting cannot be used to construct the 3 datasets. Rather, the test sets must be chronologically after the training data. The date ranges for each dataset are:

- Training dataset: 23/04/2004 – 31/12/2019 (819 observations)
- Test dataset 1: 01/01/2020 – 31/12/2023 (209 observations)
- Test dataset 2: 01/01/2024 – 31/12/2025 (105 observations)

3.2. Feature engineering

3.2.1. Stationarity analysis

Stationarity is the property of a time-series where its statistical properties and distribution are identical at all points in time. It is essential for the input features to ensure the models learn genuine predictive relationships between the input and output variables, rather than relying on the trend in the data. It also ensures distributions do not differ between the training and test sets, so models can generalise onto the test sets.

Stationarity was tested using both the Augmented Dickey-Fuller (ADF) test and the KPSS test. For the ADF test, the null hypothesis is the presence of a unit root, indicating non-stationarity. For the KPSS test, the null hypothesis is stationarity against an alternative of a unit root. Both tests suffer from low power (Arltová & Fedorová, 2016), and so by utilising both tests with their opposing null hypotheses, more confidence can be placed in the results they agree on.

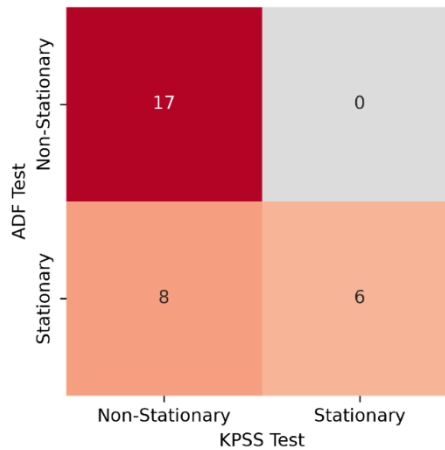


Figure 3.1: Agreement between ADF and KPSS stationarity tests, showing the number of features falling into each combination of test outcomes

Where they agreed on a time-series being non-stationary, first differencing was applied, or log-returns for price series. Where they agreed on a time-series being stationary, no transformation was applied. Where they disagreed, the tests were deemed inconclusive, and a decision was made based on domain knowledge, inspection of the time-series

graphs, and consideration of the p-values from the stationarity tests. Figure 3.1 displays the degree of agreement between the tests, showing high levels of agreement with 8 inconclusive results. The full results and the transformation applied for each feature are in Appendix C. The tests were re-applied post-transformation to check stationarity.

3.2.2. Feature selection

Due to the limited sample size, the feature set used as input data for the models needs to be shrunk significantly. This mitigates the risk of the models overfitting to the training data and ensures they are capable of learning the patterns in the data. Dimensionality reduction techniques like Principal Component Analysis are not appropriate for this study because XAI feature importance values would lose any meaningful interpretation. It is important that the impact of individual features can be cleanly distinguished. Therefore, a strict feature selection process was necessary.

First, features with minimal history were excluded. A threshold of 01/01/2005 was chosen, ensuring a sufficient minimum number of weekly observations.

Second, groups of features that provide similar information are identified and only one feature from each group is kept. The rationale is that if features are similar, then the signal within them will overlap and so some of the features will be redundant to the models. Additionally, XAI methods like SHAP assume all features are independent. When it identifies a prediction performance drop from removing a set of correlated features, it cannot determine which of the features is the true original source of the signal. Therefore, it shares the importance value between them.

To identify related features, pairwise Pearson correlation and Mutual Information were computed for all pairs of input features. Pearson correlation is defined as $\rho_{X,Y} = \frac{cov(X,Y)}{\sigma_X \sigma_Y}$. A high value indicates high linear correlation between the two features, while a low value indicates low linear correlation.

Mutual information for continuous variables is defined as:

$$\begin{aligned} I(X; Y) &= H(X) - H(X|Y) = H(Y) - H(Y|X), \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left[f(x, y) \log_2 \frac{f(x, y)}{f(x)f(y)} \right] dx dy, \end{aligned}$$

where $H(X)$ is the marginal entropy of X , $H(X|Y)$ is the conditional entropy of X given Y , $f(x)$ is the marginal probability density function of x and $f(x,y)$ is the joint probability density function of x and y . Mutual information is able to measure non-linear relationships as well as linear relationships. It is computed using scikit-learn's `mutual_info_regression` function, which estimated $I(X;Y)$ using a k-nearest-neighbour estimator.

The pairwise correlations and mutual information values were each aggregated into matrices and composed into graph structures where the edge weights are the pairwise

correlation/mutual information values. Clusters of similar variables were then identified by applying a hierarchical clustering algorithm. One feature was kept from each cluster.

Finally, correlation and mutual information were calculated between each lagged feature and the target variable. Features with low correlation and mutual information were removed, with consideration given to maintaining a diverse set of feature types to allow the XAI analysis in Experiment 1 to assess trends in which types of features are most and least important. This resulted in 16 features remaining, which was deemed appropriate for the model types and dataset size in this study. The final set of features included are shown in Table 3.1.

Table 3.1 Features provided as inputs to the forecasting models

Bloomberg ID	Description
VIX Index	VIX Index
GT10 Govt	10Y Treasury Yield
NAPMPMI Index	US Purchasing Managers Index
CL1 Comdty	WTI Price
CRK321M1 Index	321 Crack Spread
DSUPKERO Index	Kerosene Days of Supply
DSUPGASL Index	Gasoline Days of Supply
BAKETOT Index	Active US Oil & Gas Rigs
BAKEGAS Index	Active US Gas Rigs
DOESCRUD Index	US Crude Inventory
DOESCROK Index	Cushing Crude Inventory
DOEPGRIN Index	US Crude Processed Volume
DOEPPERC Index	US Refinery Utilization Rate
IP Index	US Industrial Production
EPUSTOT Index	US Electricity Output
CL1 H/L	WTI High/Low

3.2.3. Scaling

The LSTM forecasting model requires the input features matrix to be rescaled for optimal performance. Without rescaling, features whose values are of greater magnitude will become the primary focus when model weights are updated during training passes, since they would have the largest impact on node values and therefore error values. Having all features be at the same scale also speeds up convergence.

The rescaling is done using the StandardScaler function from the scikit-learn library. It rescales values according to the formula:

$$x' = \frac{x - \mu_x}{\sigma_x}.$$

μ_x and σ_x are computed from only the training dataset to avoid data leakage, and then used to rescale all the datasets.

3.3. Model Validation Framework

K-fold cross validation is used for the purposes of hyperparameter optimisation via grid search. Cross validation avoids the need for a separate validation held-out dataset, preserving all the observations in our limited sample size for the training and test sets. It involves generating multiple different training and validation splits from the original training data. The model is then trained on each training set and evaluated on the corresponding validation set, with the mean result for the chosen evaluation metric taken as the final reading. By computing this across many different combinations of model parameters, the parameter set that delivers the best performance can be identified without any risk of data leakage. The hyperparameters tested for each model are detailed in Section 3.4.1.

Since observations in a time series dataset are not independent from earlier time steps, random sampling cannot be used to generate each test and validation allocation for cross validation. Training observations must come chronologically prior to validation observations to avoid lookahead bias. Therefore, a time-aware cross validation approach is necessary. The expanding window approach is selected for this study. This involves splitting the dataset into $K+1$ contiguous time periods. The model is trained on observations within period 1 to period $k-1$ inclusive and then evaluated on data from period k , for all k from 2 to $K+1$ inclusive. The chosen evaluation metric is computed each time and the mean of these results taken as the final reading. $K = 5$ is used in this study.

3.4. Implementation

All analysis is conducted using Python. The following sections detail the implementation of the models and XAI methods, including the external libraries used.

3.4.1. Implementation of forecasting models

The XGBoost model is implemented using the XGBoost library, with a learning rate of 0.05, subsample of 0.8, and column sample by tree of 0.8.

The LSTM model was implemented using the PyTorch library with a dropout rate of 0.2, batch size of 32, learning rate of 0.001, maximum epochs of 100, and patience of 15.

The TabPFN model was implemented using the TabPFN Regressor model from the TabPFN library. By design, it has no parameters to set, although the fit mode was set to “fit_with_cache” to speed up prediction calls. This mode performs attention within the training set during training (fitting), so it does not need to be performed for every prediction call.

As well as the set parameters described above, the XGBoost and LSTM models had hyperparameters that were chosen via grid search to maximise directional accuracy, using the time-aware cross validation framework described in Section 3.3. These

hyperparameters are listed in Table 3.2, along with the tested values and the final value chosen.

Table 3.2 Model hyperparameters optimised via grid search

Model	Hyperparameter	Tested Values	Optimal Value
XGBoost	Number of estimators	100, 150, 200	100
	Maximum tree depth	3, 4, 5	3
LSTM	Number of layers	2, 3	2
	Hidden layer size	8, 16, 32	32
	Number of lags	6, 8	8

The performance of each model on the first test set is presented in Table 3.3. All three models perform within the expected range of 54-60% directional accuracy – in line with prior literature, and reasonable given this is a weekly returns forecasting task in a noisy environment. P-tests were conducted on the directional accuracy of each model using a binomial test and a null hypothesis equal to the proportion of positive returns in test set 1 (the accuracy a naïve always-long classifier would achieve). These show both the XGBoost and LSTM models are statistically significant at a 10% level, and the XGBoost model is at the 5% level, with the results unlikely to have occurred from random chance. The XGBoost model achieves the highest directional accuracy, corroborating what previous studies have found. The LSTM model also has good accuracy. It is worth noting that all 3 models exhibit prediction bias in favour of positive returns, despite the training dataset being well-balanced between positive and negative returns.

Table 3.3 Performance summary for all three forecasting models

Metric	XGBoost	LSTM	TabPFN
Directional Accuracy (%)	58.37	56.93	54.55
Directional Accuracy p-value	0.03	0.09	0.22
MSE ($\times 10^{-3}$)	2.00	2.03	1.97
MAE ($\times 10^{-2}$)	3.36	3.33	3.34
ROC-AUC	0.534	0.537	0.518

3.4.2. Implementation of XAI methods

SHAP explanations were computed for the XGBoost and LSTM models using the standard KernelSHAP library with the number of samples parameter set to 500. The background data was generated using the built-in k-means function with 50 datapoints used. For the TabPFN model, the purpose-built imputation-based shapiq explainer was used, from the TabPFN extensions interpretability library. This is a more efficient and scalable implementation, which was necessary for the TabPFN model.

LIME explanations were computed using the LIME library for all models.

WinIT explanations were computed using a self-built implementation based on Islam & Fox's (2025) adaptation of the original WinIT algorithm.

All three XAI methods were applied with a random state of 42, set prior to the calculation of each set of explanations. This ensured reproducibility of the explanations, since otherwise the random sampling present in all three methods could produce slight variation between runs.

Chapter 4

Exp. 1: Comparison of Explanations Across Methods and Models

This chapter presents Experiment 1, assessing which features each model identifies as globally important, whether different ML models and XAI methods agree on these rankings, and how faithful and stable these explanations are. It starts with an introduction and motivation, followed by a description of the experimental design, including the definitions and implementations of the analytical metrics used for rank correlation, faithfulness, and stability. It then presents the results of the experiment, before concluding on the key findings.

4.1. Introduction

XAI explanations are often used by practitioners to provide insight into a model's decision, understand why certain predictions are made, and to measure feature importance. However, most applications assume the reliability and trustworthiness of the chosen XAI methods. Existing research indicates that not all XAI methods are equal, and they can perform differently in different contexts (e.g. for different forecasting models or on different datasets). With this in mind, any research that applies XAI methods should first evaluate them and compare multiple methods before deciding to rely on the explanations produced by them. It also raises some important questions for practitioners: which methods are most trustworthy, and how should trustworthiness even be measured? This is the core investigation of this experiment.

This experiment aims to assess the globally aggregated properties of all three XAI methods across all three models. It first identifies which features the forecasting models rely on to forecast energy indices, before focussing on how trustworthy these explanations are. The latter part is assessed by analysing whether methods and models converge on a consistent view of feature importance and also by formally evaluating the faithfulness and stability of explanations.

The remainder of this chapter proceeds as follows: Section 4.2 details the experiment design, including the statistical methodologies used and the specific faithfulness and stability metrics implemented; Section 4.3 presents the results of the experiment; and Section 4.4 discusses the implications of these findings in the context overall aims of this experiment.

4.2. Experiment Design

The aim of this experiment is to assess the global properties of all three XAI methods across all three models. It attempts to answer three key questions: (1) which features do different models rely on, (2) do different models and explanation methods converge on a consistent view of feature importance, and (3) how trustworthy are these explanations? Together, these questions give a comprehensive view on how these XAI methods perform within the energy price forecasting context, and provide practitioners concrete insights to aid them in explaining ML prediction models.

First, global feature attributions are computed for each applicable model-method pair: SHAP and LIME across all three models (XGBoost, LSTM, and TabPFN) and WinIT restricted to the LSTM (reflecting its design as a sequence-specific attribution method incompatible with the non-sequential architectures). Global feature importance rankings are presented for all method-model pairs, before further examination is then undertaken, including the analysis of cross-method and cross-model feature rank correlations, and evaluation of faithfulness and stability over all method-model pairs. The details of these analyses are described below. All analysis is conducted on test set 1.

4.2.1. Global feature importance and Spearman correlation

The analysis begins by presenting the feature importance rankings for all method-model pairs. These are assessed with consideration for the existing literature on important factors for energy forecasting and economic intuition. Lag importance profiles for the LSTM model (the only model with multi-lag input data) are also analysed.

Cross-method and cross-model correlations between feature attributions are then presented to assess the consistency of explanations across methods and models. For this, Spearman rank correlation is used to calculate correlations between feature importance rankings. It is more appropriate than Pearson correlation of feature importance values would be for this analysis, because feature importance values are not comparable between different XAI methods. Therefore, feature importance *rankings* are the appropriate and comparable results across all explanations, and Spearman correlation is the established method to compare similarity between rankings. Spearman correlation is defined as:

$$\rho = \frac{\text{cov}(\text{rank}(X), \text{rank}(Y))}{\sigma_{\text{rank}(x)} \sigma_{\text{rank}(y)}}.$$

When there are no tied ranks, this calculation can be simplified to:

$$\rho = 1 - \frac{6 \sum_{i=1}^n (\text{rank}(X_i) - \text{rank}(Y_i))^2}{n(n^2 - 1)}.$$

Values close to +1 indicate strong agreement on feature importance orderings, negative values indicate divergent explanations (with -1 indicating completely opposing orderings), and values close to 0 indicate a lack of clear agreement or disagreement.

4.2.2. Faithfulness correlation

Faithfulness is measured using faithfulness correlation (Bhatt, et al., 2020). This measures the extent to which per-prediction feature importance rankings from an XAI method align with the change in model output when those features are occluded from the model.

Given a prediction model f , an explanation function g , a datapoint x , and a subset size $|S|$, the faithfulness of g to f at x is defined as:

$$\mu_F(f, g; x) = \text{corr}_{S \in \binom{[d]}{|S|}} \left(\sum_{i \in S} g(f, x)_i, f(x) - f(x_{[x_S = \bar{x}_S]}) \right),$$

where x_s is the sub-vector of input features corresponding to the feature subset S , and $\bar{x}_s$ denotes where x_s is set to a baseline (in this study, the feature means are used as the baseline). This is computed by fixing $|S|$ and randomly sampling for different subsets S to estimate the above correlation.

Essentially, it is computed by obscuring random subsets S of the feature set where the size of the subsets $|S|$ is fixed, replacing those feature values with their feature means, and then evaluating the model on this perturbed input data. After repeatedly sampling this for different subsets of features, the Pearson correlation is computed between the changes in model outputs (relative to the original output when no inputs were occluded) and the sum of feature importance values for each corresponding subset of features.

Values range between -1 and +1, with values close to +1 indicating feature importance values strongly reflect true impact on model output. Negative values indicate feature importance values that are opposite to actual feature impact.

This study uses the implementation of faithfulness correlation from the Quantus python package (Hedström, et al., 2023). Due to the requirements of this package, the XGBoost and TabPFN models had to be wrapped in PyTorch modules (the LSTM model was already a PyTorch model).

4.2.3. Local Lipschitz stability

Stability is measured using the local Lipschitz estimate (Alvarez-Melis & Jaakkola, 2018). This measures the smoothness of the explanation function, assessing whether it produces similar attributions for similar (neighbouring) inputs. This formalises the intuition that an explanation should not change substantially in response to an infinitesimal, prediction-preserving perturbation of the input.

Given an explanation function g , a datapoint x , and a neighbourhood of similar inputs from the dataset $\mathcal{N}(x) = \{x' \in X: \|x - x'\|_2 \leq \varepsilon\}$ defined by a radius ε , the local Lipschitz estimate at x is defined as:

$$\hat{L}(x_i) = \max_{x_j \in \mathcal{N}(x_i)} \frac{\|g(x_i) - g(x_j)\|_2}{\|x_i - x_j\|_2}.$$

This is computed by repeatedly sampling perturbed points x' within the ε -ball around x , evaluating the explanation function on each, and taking the maximum ratio of the change in attribution to the change in input across all sampled perturbations.

Unlike faithfulness correlation, local Lipschitz estimates are not bounded. Values are non-negative, with lower values indicating that the explanation changes smoothly with small input perturbations (i.e. greater stability), and higher values indicating that similar inputs receive disproportionately different explanations, undermining trust in the explanation.

This study uses the implementation of local Lipschitz estimate from the Quantus python package (Hedström, et al., 2023).

4.3. Results

4.3.1. Global feature importance

Table 4.1 reports the 5 most and 5 least important features for each model-method pair at a global level, evaluated on test set 1. There are several similarities in the rankings, indicating some consistency in the signals each model is able to identify and exploit within the data.

Table 4.1: Top 5 and bottom 5 features by feature importance ranking across models

Rank	XGBoost SHAP	XGBoost LIME	LSTM SHAP	LSTM LIME	LSTM WinIT	TabPFN SHAP	TabPFN LIME
1	DOESCRUD	DOESCRUD	CL1 H/L	BAKEGAS	DSUPKERO	DOESCRUD	DOESCRUD
2	DSUPGASL	CL1	DOESCRUD	CL1 H/L	CL1 H/L	DSUPGASL	DSUPGASL
3	CL1	DSUPGASL	CRK321M1	CRK321M1	NAPMPMI	DOEPPERC	DOEPPERC
4	CRK321M1	DOEPGRIN	DSUPGASL	DOEPGRIN	DSUPGASL	DSUPKERO	VIX
5	DOEPPERC	GT10	IP	IP	DOEPPERC	CL1 H/L	CL1
12	BAKETOT	CL1 H/L	GT10	BAKETOT	CL1	DOESCROK	GT10
13	BAKEGAS	BAKETOT	EPUSTOT	EPUSTOT	EPUSTOT	DOEPGRIN	DOEPGRIN
14	IP	BAKEGAS	VIX	DSUPKERO	GT10	BAKETOT	EPUSTOT
15	NAPMPMI	DOESCROK	BAKETOT	CL1	VIX	GT10	IP
16	DOESCROK	IP	BAKEGAS	VIX	BAKETOT	BAKEGAS	BAKETOT

The DOESCRUD index, a measure of US crude oil inventory stocks, is consistently amongst the top features, ranking first for the non-temporal models, as well as for the LSTM under SHAP. This aligns with standard economic theory, as crude oil inventory is a strong indicator of short-term oil supply. Similarly, DSUPGASL – another supply side indicator measuring the days of supply for gasoline – ranks 2-4 under all method-model pairs except LSTM-LIME. In contrast, both the BAKEGAS and BAKETOT indices – which measure the number of active US gas rigs and the number of active US oil and gas rigs respectively – tend to rank low for all the models. Despite being popular measures of oil supply, they don’t appear to provide any useful signal for weekly forecasting here – or at least any additional signal beyond what the other features provide.

The LSTM model relies heavily on the previous week’s WTI high-over-low (CL1 H/L) – ranking it first or second across all XAI methods – while the XGBoost and TabPFN models rank this feature much lower. A similar, but slightly weaker, effect can be seen for the CRK321M1 Index, which is also consistently ranked relatively higher for the LSTM than for the other models. The mechanism for this can be found by inspecting the importance of each feature per lag for the LSTM, displayed for SHAP in Figure 4.1. Both CL1 H/L and CRK321M1 Index show high importance for earlier lags, with attribution plots peaking at lag 2 for CL1 H/L and lag 3 for the CRK321M1 Index. In contrast, most other features show a monotonic increase from lag 8 to lag 1, with lag 1 carrying most of the signal. This suggests that CL1 H/L and CRK321M1 Index’s higher relative importance for the LSTM is due to the additional temporal information the LSTM is able to extract from its input sequence, which the non-temporal models are unable to see. Such examples demonstrate the strength of XAI methods for assessing how suitable an ML model is for a given forecasting task, and whether the temporal nature of the data is providing any benefit.

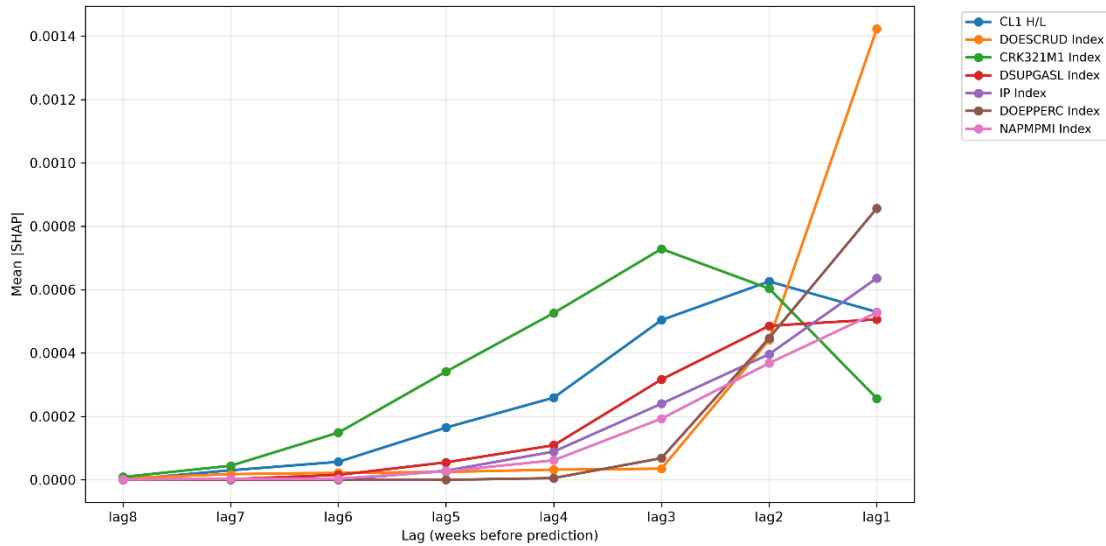


Figure 4.1: Global lag importance profiles by feature for the LSTM model under SHAP, for the top 8 features

Figure 4.2 shows the aggregated lag importance profiles for each XAI method on the LSTM model. Despite some features displaying non-monotonic lag attribution profiles, aggregated lag profiles for both SHAP and WinIT take the expected monotonic shape, increasing in importance towards lag 1. This corroborates existing literature and economic intuition, since more recent lags are expected to carry more relevant information. However, the lag profile of LIME is counter-intuitive, collapsing unrealistically at lags 3 and 4.

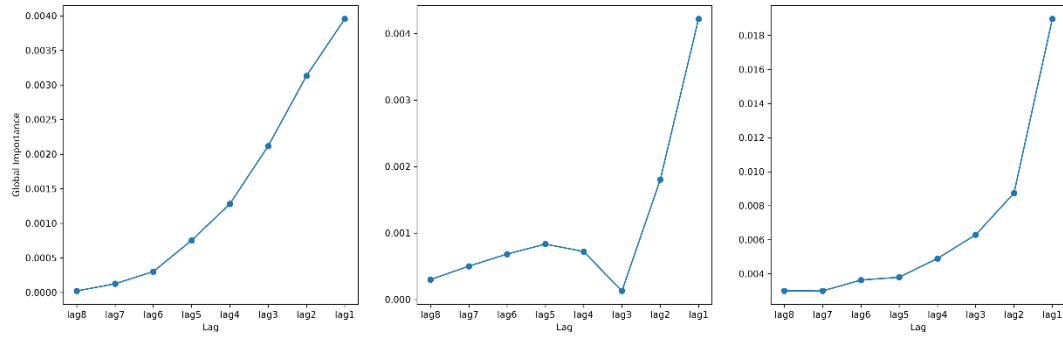


Figure 4.2: Global lag importance profiles for the LSTM model under SHAP (left), LIME (middle) and WinIT (right)

Figure 4.3 displays the correlation in feature rankings between each method-model pair. Immediately, the anomalous nature of LIME on the LSTM becomes clear – it has low correlation with the other LSTM results, and even negative correlation with the other models. This is clearly not a consequence of the LSTM model or LIME on their own, since the LSTM’s SHAP and WinIT explanations are positively correlated with the other model’s explanations, and so are LIME’s explanations on the other models. It is possible that this is a case of LIME struggling under high feature counts and multicollinearity, since the LSTM model’s input data has 16 features $\times$ 8 lags, significantly more than the other models which only have 1 lag. If so, it would suggest that LIME is more sensitive to high feature counts than SHAP or WinIT, and not suitable for temporal data.

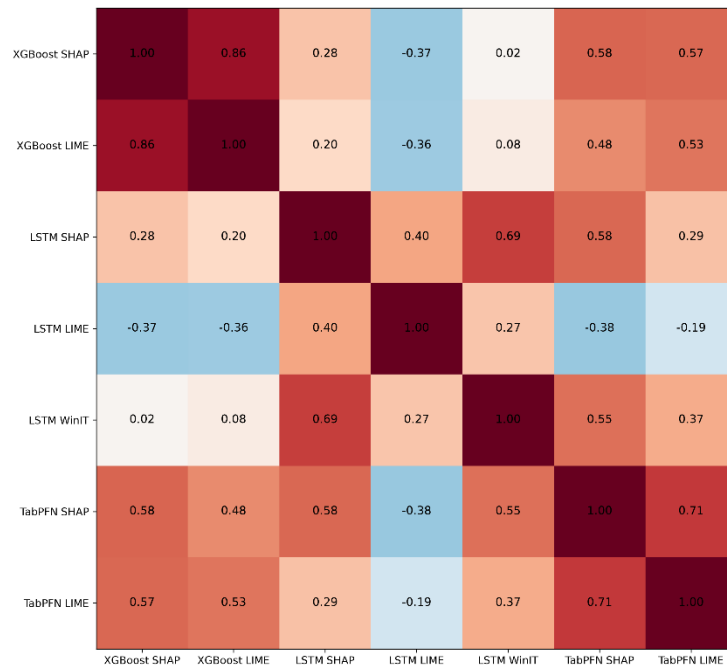


Figure 4.3: Spearman rank correlation of global feature importance rankings across all model-method pairs

Except for LSTM-LIME, we see strong cross-method agreement for each model at this global level, providing confidence in these explanations.

Interestingly, the two high-performance models (XGBoost and LSTM) have low cross-model agreement, with both actually correlating more with TabPFN (despite its poor prediction performance) than they do with each other. This suggests that, despite both the XGBoost and LSTM models successfully exploiting signal in the dataset, the signals each rely on are distinct and not highly correlated. They appear to be making their decisions based on different patterns in the data. This demonstrated another valuable use-case for XAI analysis: deducing whether individual models are homogenous or heterogenous in their decision-making – useful for assessing the overall riskiness of an ensemble model.

4.3.2. Faithfulness

Table 4.2 presents the mean faithfulness correlation for each method-model pair. The SHAP explanations consistently display a high degree of faithfulness across all models. The highest faithfulness is achieved on the TabPFN model (0.913), while it is lowest for the LSTM model. The latter result is likely due to the temporal nature of the input data, where adjacent lags of a feature are likely to be autocorrelated, since a documented weakness of SHAP is in its ability to allocate importance between correlated features. This is evidence that SHAP should be used cautiously when temporal data is involved.

Table 4.2: Faithfulness correlation values for all method-model pairs

XAI Method	XGBoost	LSTM	TabPFN
SHAP	0.808	0.734	0.913
LIME	0.022	-0.071	0.020
WinIT	n.a.	0.006	n.a.

LIME and WinIT show extremely poor faithfulness – barely above random. LIME even displays a slightly negative faithfulness correlation for the LSTM model, indicating it is showing the opposite of the truth. This means that, at the local per-prediction level, only SHAP feature importance values are reliable indicators of the improvement in model performance when those features are provided to the model, while LIME and WinIT’s importance values are not.

However, this does not mean LIME and WinIT are not useful. Firstly, as discussed in Section 4.3.1., these XAI methods produce similar rankings to the SHAP rankings at a global level (with the exception of LIME on the LSTM). This suggests that while they struggle to allocate importance at a per-prediction local level (which is what faithfulness correlation is based on), this error appears to cancel out when aggregated to a global level.

Furthermore, in the case of WinIT, a low faithfulness correlation should not be taken to mean the explanations are wrong, but rather that caution must be taken with respect to their interpretation. WinIT’s expanding window obscuring method is specifically

designed to expose the marginal impact of removing an additional time lag of a feature, given all more recent time lags are already obscured. This marginal impact should not be misinterpreted as the time lag’s overall importance to the model, which is what faithfulness correlation is comparing it to. In general, it is clear that not all XAI methods are designed to produce the same results, and caution must be taken to interpret them within the context they were designed for.

For LIME on the LSTM specifically, the negative faithfulness result provides evidence that its out-of-the-norm feature and lag rankings discussed in 4.3.1 – where, for example, BAKEGAS ranked top and lag 3 collapsed in importance – were not truly representative of model behaviour. This highlights the risks of blindly relying on a single XAI method, and also the importance of evaluating faithfulness for each specific XAI use-case.

4.3.3. Stability

Table 4.3 displays the results of the Local Lipschitz stability test. Lower values indicate greater stability (less deviation under small input perturbation). Unlike with the faithfulness tests, SHAP does not strictly dominate the other methods here. It does outperform LIME on both non-temporal models (XGBoost and TabPFN), with the biggest difference for TabPFN, but for the LSTM model, this relationship flips, with LIME significantly more stable than SHAP. This reflects a similar trend to what was seen in the faithfulness analysis – SHAP performs worse on the LSTM than the other models. However, LIME’s apparent stability on the LSTM should be read alongside its near-zero faithfulness on that model – an explainer that is not tracking the model cannot be meaningfully perturbed.

Table 4.3 Local Lipschitz stability values for all method-model pairs

XAI Method	XGBoost	LSTM	TabPFN
SHAP	0.919	1.506	0.278
LIME	1.052	0.503	1.028
WinIT	n.a.	1.336	n.a.

WinIT performs similarly poorly to SHAP on the LSTM. However, this might not be measuring the same instability as for the other methods. Rather than being the direct effects of small perturbations in the inputs, it is likely that this is actually a consequence of a weakness in the design of the WinIT algorithm, which is highly sensitive to random sampling. When each input value is occluded in WinIT, the replacement value is randomly selected from the full time-series of that feature-lag, and the change in prediction is recorded. However, this random selection is only performed once, rather than a full random sampling happening, so the change in model performance is highly dependent on which single replacement value was selected. This has the potential to create large changes in feature attributions during separate runs of the WinIT algorithm.

SHAP-TabPFN is the most stable and most faithful of all the method-model pairs despite TabPFN having the weakest out-of-sample prediction performance, so it is clear that

strong XAI performance does not correlate with prediction performance. It is possible that weaker predictive performance actually makes the stability metrics appear better, since a model that has struggled to exploit patterns in the data might not have its output vary much under perturbation to its inputs. Therefore, we should not place too much weight on these results, and it is important that XAI evaluation metrics do not overshadow standard predictive performance evaluation as a first-stage assessment of model suitability.

4.4. Discussion

This experiment analysed the out-of-sample feature importance and lag importance values produced by 3 different XAI methods on 3 different model architectures. By comparing explanations across models and methods, this experiment demonstrated strong alignment across most model-methods pairs, with an exception for LIME on the LSTM model. The most important features from across the models were short-term supply side indicators. Interestingly the long-term supply side indicators of the BAKEGAS and BAKETOT indices, which are standard in the industry, were not important to the model’s predictions at all. XAI analysis on the LSTM model also highlighted where the temporal input data was providing benefit to the model.

LIME on the LSTM produced implausible results across both features and lags, demonstrating the importance to practitioners of evaluating XAI methods in each specific use-case, and comparing explanations from multiple methods, rather than blindly trusting the explanations from a single method.

The faithfulness evaluation of the XAI methods delivered several significant findings. Firstly, SHAP was consistently more faithful than LIME, despite the strong feature ranking agreement at the global level. Secondly, both LIME and SHAP performed worse on the LSTM than the other models. This second result aligns with existing research given the autocorrelation present between adjacent lags in the temporal input data of the LSTM. It highlights that caution must be taken when using these XAI methods on time-series data.

The faithfulness and stability XAI evaluation also demonstrated that current XAI methods do not always tell the full story, and must be interpreted with caution. WinIT had very poor faithfulness correlation, but this is expected given it is designed to measure each lag’s marginal importance rather than their overall importance like SHAP is. Additionally, TabPFN-SHAP achieved the best stability results, but this is likely a consequence of its poor predictive performance and so does not mean the model or its XAI explanations are necessarily useful.

Chapter 5

Exp. 2: Local Attribution Patterns Across Prediction Outcomes

This chapter presents Experiment 2, examining how local explanations vary across trading outcomes (profitable vs. loss-making). The introduction motivates the topic, which is followed by a description of the experimental design. The results are then presented, and then the key findings are discussed.

5.1. Introduction

Existing literature has found that XAI explanations can shift significantly between periods of high and low model performance, and can produce identifiable signals of model failure. However, this analysis has typically been purely qualitative, lacking the statistical rigour that would enable confident use of these insights. Furthermore, no research has investigated whether systematic shifts can identify model failure at the more granular level of individual predictions, to discriminate between correct and incorrect predictions.

Understanding whether explanations carry diagnostic information about the model's reliability on a given prediction would be extremely valuable. If they do, then this would offer a mechanism for a model to flag its own low-confidence predictions in real time, providing opportunities to operationalise these signals in live trading strategies. If not, then this in itself would be an important finding, providing more clarity on the true extent of the systematic shifts identified in previous literature. It would also suggest caution should be taken by practitioners using explanations as indicators of model reliability, as some previous research has suggested.

Therefore, the aim of this chapter is to test whether local attribution patterns discriminate correct from incorrect predictions. This is assessed by comparing the distributions of feature importance attributions for correct and incorrect predictions using divergence and entropy measures – across both feature and time-lag dimensions – and by measuring changes in cross-method and cross-model agreement.

This question also sits at the centre of this thesis, being essential for the operationalisation stage in Experiment 3. No specific confidence signal is assumed for this study, but rather, a variety of plausible attribution patterns are tested to determine which attribution patterns – if any – carry a discriminating signal. Whichever patterns

prove discriminating here are what Experiment 3 operationalises as a confidence gate. The result of this chapter therefore directly determines the direction of Experiment 3.

The remainder of this chapter is organised as follows: Section 5.2 details the design of this experiment, including the statistical methodologies used; Section 5.3 presents the results of this experiment; and Section 5.4 discusses the implications of the findings in the context of the overarching aim of the experiment.

5.2. Experiment Design

Experiment 2 aims to assess whether local XAI attribution patterns can systematically discriminate between a model's correct and incorrect predictions. This is tested across all forecasting models except TabPFN, and across both SHAP and LIME, with WinIT excluded due to scope constraints and its high instability between runs. All analysis is conducted on test set 1.

The TabPFN model was excluded due to its poor predictive performance, which was indistinguishable from random even at the 10% significance level. Attribution analysis presupposes that a model's predictions encode a genuine, learned decision process, and this experiment then asks whether that process leaves a detectable attribution signature under error. If a model's directional accuracy is statistically indistinguishable from random, its "correct" and "incorrect" prediction groups would not be two meaningfully different decision regimes, but would instead be closer to a random split. Any discriminating signal found would more plausibly reflect noise in the attribution estimation process than a genuine error-correlated shift in explanation behaviour. Excluding TabPFN keeps the discriminability tests confined to models whose predictions are worth explaining in the first place.

This experiment tests three potential discriminating signals in turn, corresponding to the following three hypotheses: (1) do feature importance distribution differ between correct and incorrect predictions, (2) do the entropies of feature importance distributions differ between correct and incorrect predictions, and (3), is cross-method or cross-model agreement higher for correct prediction than for incorrect predictions? For the first two hypotheses, this is tested across both feature and lag dimensions for the LSTM model.

The first hypothesis is tested for each applicable model-method pair by partitioning test set 1 predictions into directionally correct and incorrect groups, computing Jensen-Shannon divergence (JSD) between the feature (and lag) importance distributions of the two groups, and assessing statistical significance via a permutation test.

The second hypothesis is tested by computing per-prediction Shannon entropies of the feature importance distributions (separately for both feature and lag dimensions). This is tested for association with prediction outcome in two ways: (i) a binary test, comparing entropy between correct and incorrect groups via the Mann-Whitney U test; and (ii) a

continuous test, computing the Spearman correlation between per-instance attribution entropy and absolute prediction error. Both tests yield a p-value.

The third hypothesis is examined at two levels: (i) globally – comparing the aggregate correct-group correlation against the aggregate incorrect-group correlation descriptively (no significance test applied); and (ii) per-prediction – computing a cross-method correlation for each individual prediction and testing whether this per-prediction correlation differs between correct and incorrect groups via the Mann-Whitney U test.

The statistical techniques used to test these hypotheses are detailed below.

5.2.1. Jensen-Shannon divergence

Jensen-Shannon divergence (JSD) is a measure of the difference between two distributions. It is bounded between 0 and 1 (using the base-2 logarithm), symmetric, and well-defined even where P or Q assigns zero mass to a feature or lag, making it preferable to simple KL divergence for this comparison. A value of 0 indicates the distributions are identical while a value of 1 indicates there is no overlap. It is used in this experiment to measure the difference in the feature (and lag) importance distributions of the correct and incorrect predictions.

For each method-model pair under investigation, the full array of test set 1 predictions is first separated into directionally correct and incorrect prediction subsets. For each subset, aggregate feature importance is computed by calculating the mean importance value per feature (or lag) across all the predictions in that subset. The resulting feature importance distributions ϕ are normalised to sum to 1:

$$\tilde{\phi}_j = \frac{\phi_j}{\sum_{j=1}^M \phi_j},$$

where M is the number of features.

The JSD value is then computed as:

$$JSD(C \parallel I) = \frac{1}{2} KL(C \parallel M) + \frac{1}{2} KL(I \parallel M),$$

where C and I are the normalised feature importance distributions for the correct and incorrect subsets respectively, M is the equally-weighted mixed distribution, and $KL(\cdot \parallel \cdot)$ denotes the Kullback-Leibler divergence, which is defined as:

$$KL(P \parallel Q) = \sum_{x \in X} P(x) \log \frac{P(x)}{Q(x)},$$

for two distributions P and Q.

Statistical significance of the observed JSD is assessed via a permutation test. The full set of predictions is randomly sampled into two groups of equal size to the correct/incorrect subsets, and the JSD between the two groups is computed. This is repeated for 1000 permutations to construct an empirical null distribution of JSD values under the

assumption of no true difference between groups. The empirical p-value is the proportion of permuted JSD values greater than or equal to the observed JSD. A significant result indicates that the attribution distributions of correct and incorrect predictions differ by more than would be expected under random label assignment.

5.2.2. Shannon entropy

Attribution concentration is measured using Shannon entropy, computed per-prediction over the normalised feature (or lag) attribution distribution:

$$H(\phi_i) = - \sum_{j=1}^M \phi_{i,j} \log \phi_{i,j},$$

where $\phi_{i,j}$ is the (normalised, non-negative) attribution assigned to feature or lag j for observation i .

Higher entropy indicates attribution is spread diffusely across features/lags, while lower entropy indicates attribution is concentrated on a small subset. The computed entropies are tested for association with prediction outcomes in two ways: (i) binary by comparing entropies for correct/incorrect subsets via an MW test, and (ii) continuously by measuring Spearman correlation with absolute error.

5.2.3. Mann-Whitney U test

The Mann-Whitney U (MW) test is a non-parametric test for whether two independent samples are drawn from the same distribution. It is preferred over a t-test or Kolmogorov-Smirnov test as it does not assume normality and is specifically suited to detecting rank shifts between groups.

The test statistic U is defined as:

$$U = \min \left[n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1, n_1 n_2 + \frac{n_2(n_2 + 1)}{2} - R_2 \right],$$

where n_1 and n_2 are the number of observations in each of the first and second groups respectively, and R_1 and R_2 are the sum of the ranks in each group when all observations from both groups are combined and ranked in ascending order from smallest to largest.

Cliff's delta is reported alongside the Mann-Whitney result as a non-parametric effect size:

$$\delta = \frac{1}{n_1 n_2} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} [\text{sign}(C_i - I_j)],$$

where C and I are the correct and incorrect prediction groups respectively.

Cliff's delta is bounded between -1 and +1. Values near 0 indicate substantial overlap between the two groups' distributions; values near ± 1 indicate complete separation.

The MW test and Cliff’s delta are utilised twice in this experiment. First, they are used to test hypothesis (2), for the binary analysis of entropy as a discriminator. Second, they are used to test hypothesis (3), to compare per-prediction cross-method correlation values between correct and incorrect groups.

5.2.4. Spearman correlation

Spearman rank correlation is used twice in this experiment. First, Spearman correlation of per-prediction entropy with pre-prediction absolute error is used for the continuous test of hypothesis (2). Second, Spearman correlation of feature rankings is used for the global test of hypothesis (3), to measure agreement in global feature importance rankings across methods and models separately for the correct and incorrect groups, which can then be compared.

For details on Spearman rank correlation and its implementation in this study, see Section 4.2.1.

5.3. Results

5.3.1. Attribution distribution divergence

Table 5.1 displays the JSD values and corresponding p-values comparing feature attributions across correct and incorrect predictions. The XGB-LIME method-model pair shows significant divergence at the 5% level. Examining the difference in feature importances for this pair, shown in Figure 5.1, it is the DOESCRUD index and CL1 price that have the greatest difference in importance between correct and incorrect predictions, being far more important for correct predictions. These also happen to be the most important features for the XGBoost model overall. In fact, all of the top 10 features by overall importance are more important for correct predictions, and this is a large difference for all of the top 7 features by overall importance. This suggests that correct predictions rely more on the dominant patterns ingrained into the model, while incorrect predictions are more random and less cleanly linked to these core patterns.

Table 5.1: JSD values and corresponding p-values measuring the difference in global feature importance distributions between the correct and incorrect predictions, for the XGBoost and LSTM models under SHAP and LIME

Model	XAI Method	JSD	p-value
XGBoost	SHAP	0.104	0.356
	LIME	0.021	0.026
LSTM	SHAP	0.068	0.977
	LIME	0.006	0.397

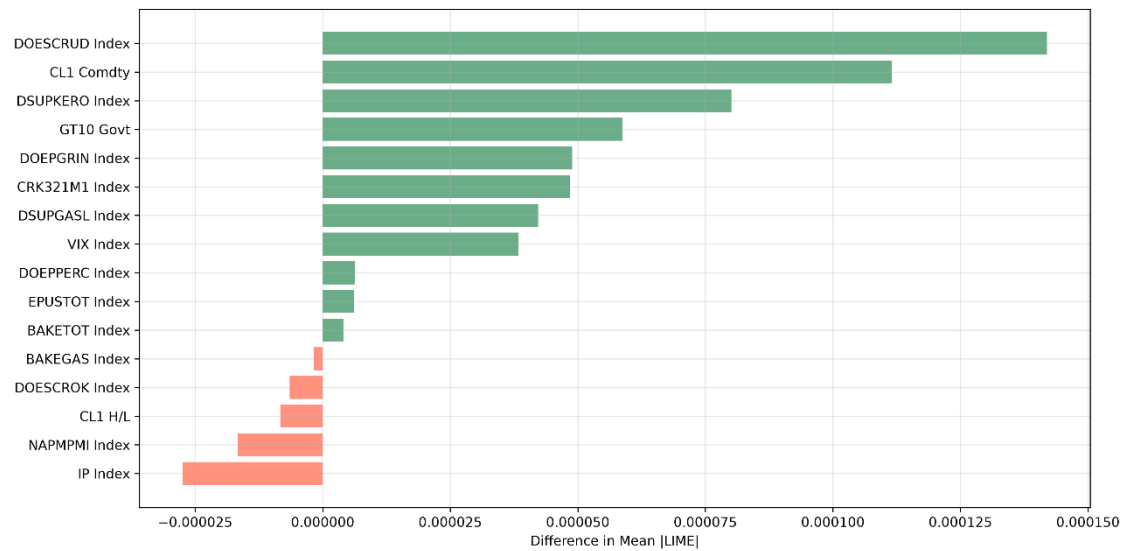


Figure 5.1: Difference in mean absolute LIME feature attribution between correct and incorrect predictions for XGBoost, ranked by magnitude of difference

Despite the significant result for XGB-LIME, overall there is no evidence this metric can consistently discriminate between correct and incorrect predictions. It appears that these models rely on the same features and feature patterns in the data across both the correct and incorrect subsets, and so incorrect predictions are not a consequence of degenerate model behaviour. Rather, it is likely that incorrect predictions stem from learned patterns not being well suited to those datapoints.

A similar result is found for lag attribution JSD divergence (LSTM only), presented in Table 5.2. The SHAP result is marginally significant at the 10% level, but overall, it is not immediately clear whether there is any discriminating power here. Further investigation into the SHAP lag profile is presented in Figure 5.2. We see that the key difference between correct and incorrect predictions is an over-reliance on lag 1 for incorrect predictions, at the expense of lags 2-4. The same pattern is not seen for the LIME lag profile, also shown in Figure 5.2, where correct and incorrect lag profiles are almost identical. This suggests that the marginally significant result for SHAP is genuinely representative of its additional discriminating power in this case.

Table 5.2: JSD values and corresponding p-values measuring the difference in global lag importance distributions between the correct and incorrect predictions, for SHAP and LIME on the LSTM model

XAI Method	JSD	p-value
SHAP	0.079	0.072
LIME	0.009	0.626

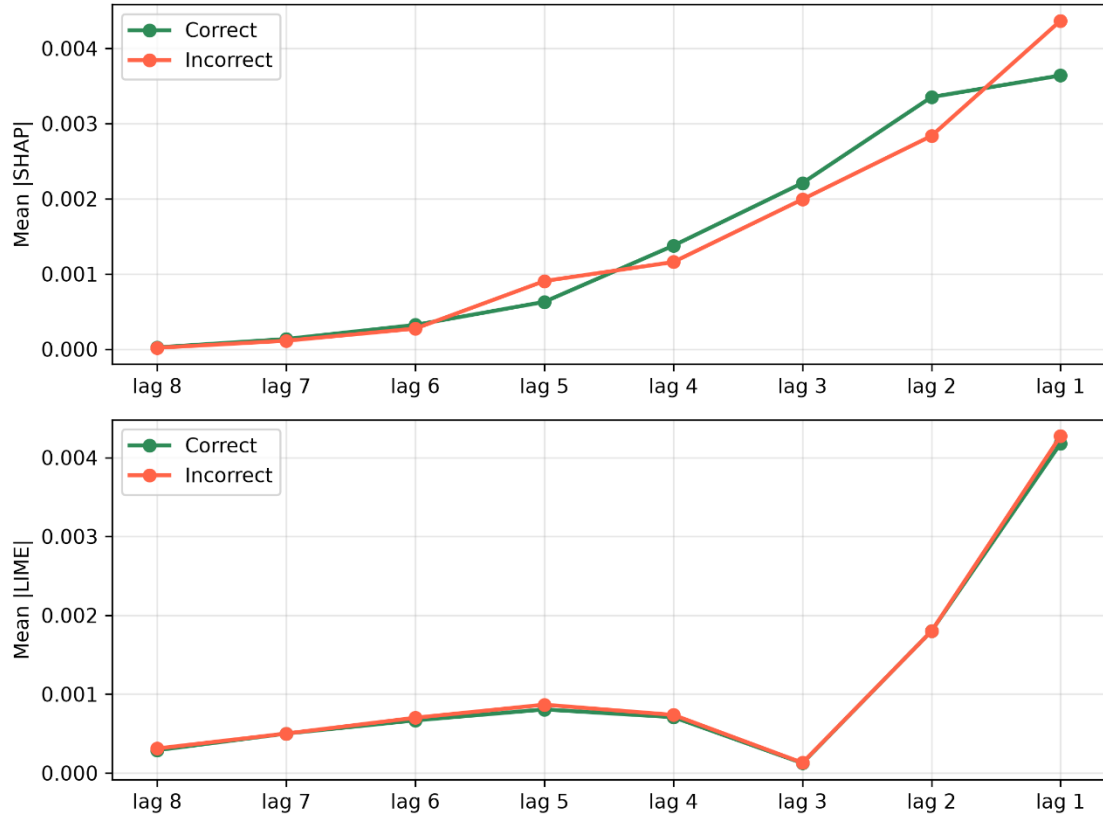


Figure 5.2: Lag importance profiles compared between correct and incorrect predictions, for SHAP (top) and LIME (bottom)

5.3.2. Attribution entropy divergence

Table 5.3 displays the results for the entropy divergence tests on feature attributions. The binary entropy divergence shows the p-value from an MW test on the entropies of the correct and incorrect subsets. The entropy correlation shows the Spearman correlation and corresponding p-value between entropy and absolute prediction error. Table 5.4 presents the same metrics for lag attributions (LSTM only).

Table 5.3 MW test p-values on feature importance entropy between correct and incorrect predictions (binary), and Spearman correlation between feature importance entropy and absolute prediction error with corresponding p-values

Model	Method	Entropy Binary test p-value	Entropy – absolute error correlation coefficient	Entropy – absolute error correlation p- value
XGB	SHAP	0.500	0.097	0.162
	LIME	0.117	0.023	0.739
LSTM	SHAP	0.737	-0.121	0.086
	LIME	0.035	0.141	0.045

Table 5.4: MW test p-values on lag importance entropy between correct and incorrect predictions (binary), and Spearman correlation between lag importance entropy and absolute prediction error with corresponding p-values

Method	Entropy Binary test p-value	Entropy – absolute error correlation coefficient	Entropy – absolute error correlation p- value
SHAP	0.653	0.107	0.128
LIME	0.203	0.160	0.023

The LSTM-LIME pair delivers the strongest results across both the feature and lag dimensions, with its explanation’s entropies appearing to be a strong discriminator for the correctness of a prediction. 3 of the 4 results are statistically significant at the 5% level, and the direction of the relationship is consistent: higher entropy indicates more error / more likely to be incorrect.

All the other model-method pairs result in marginally significant / insignificant results, suggesting this will not always be a reliable discriminator, at least on small sample sizes like those used in this experiment. It is possible that this sample size was too small to detect finer divergences, or that the specific market regimes over this period were not suited for distinguishing correct and incorrect predictions based on feature importance.

Interestingly, the LSTM-LIME pair is the strongest discriminator between correct and incorrect predictions. Yet it delivered the worst faithfulness result in Section 4.3.2. This is evidence that low faithfulness does not inhibit an XAI method from being a strong discriminator. The explanations might not accurately represent which features are most important, but the spread of importance values can still act as a strong indicator of model failure.

5.3.3. Cross model-method agreement

Figure 5.3 displays the changes in global explanation Spearman correlation from correct to incorrect predictions. Most correlations are relatively similar between the correct and incorrect subsets. Only the correlation in SHAP explanations across the two models shows any noticeable change and, in contrast to what was expected, shows less agreement between the explanations. This is further evidence that the two models are exploiting distinct and uncorrelated signals in the dataset, since the features driving their decisions are even less correlated for correct predictions than incorrect predictions.

Results for divergence in cross-method Spearman correlations at the per-prediction level are reported in Table 5.5, measuring divergence between the correct and incorrect subsets using an MW test. Again, there is little evidence of a systematic difference. Only the XGBoost cross-method agreement shows any positive result, and it is only marginally significant. The Cliff’s delta for this result was small and positive, suggesting a slight increase in cross-method agreement for correct predictions. However, there is not enough evidence to reliably determine whether this is just the result of random noise.

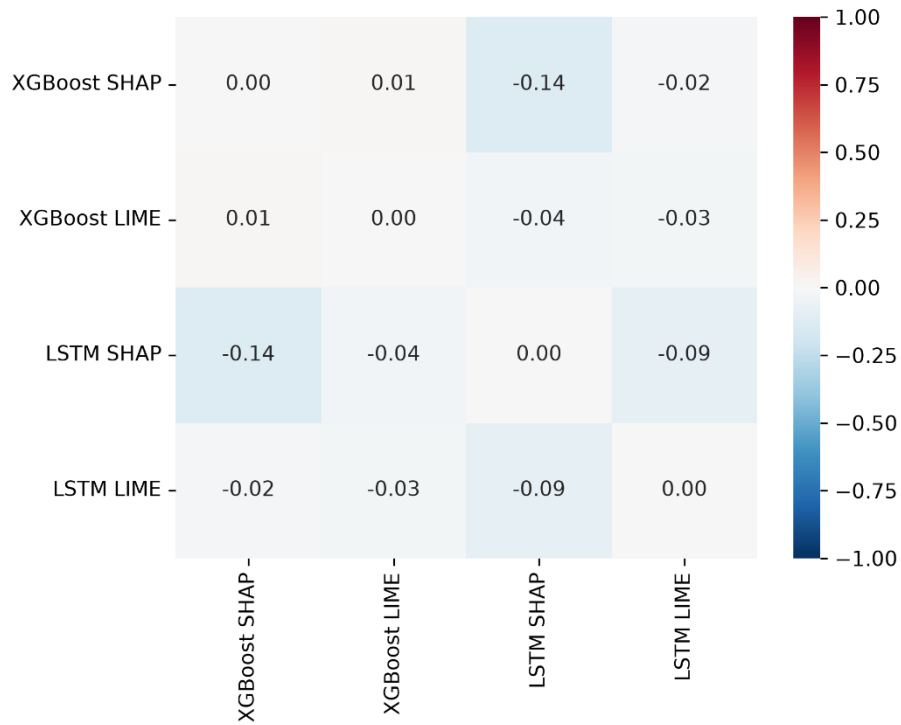


Figure 5.3: Difference in cross-method / cross-model Spearman correlation of global feature importance between correct and incorrect predictions, for the XGBoost and LSTM models under SHAP and LIME

Table 5.5 MW test p-values and Cliff’s delta comparing per-prediction SHAP-LIME agreement between correct and incorrect predictions, for the XGBoost and LSTM models

Model	Cross-method agreement		Cliff's delta
	MW test p-value		
XGBoost	0.088*		0.138
LSTM	0.921		-0.008

5.4. Discussion

This experiment assessed whether local XAI explanations could be used to systematically discriminate between correct and incorrect predictions, serving as a signal of model failure. It did so by testing three hypotheses: (1) do feature importance distributions differ between correct and incorrect predictions, (2) do the entropies of feature importance distributions differ between correct and incorrect predictions, and (3), is cross-method or cross-model agreement higher for correct prediction than for incorrect predictions?

The results present mixed support for the overarching question. Regarding hypothesis 1, there is weak support that the feature importance distributions from some XAI methods can systematically differ between correct and incorrect predictions, with LIME

feature importance for the XGBoost model and SHAP lag importances for the LSTM model delivering statistically significant differences in distributions. However, all other model-method pairs deliver insignificant results, and it is clear this cannot be relied on as a consistent signal of model failure.

Hypothesis 2 is better supported, with the entropy of LIME attributions on the LSTM consistently generating strong discriminating signals across both feature and lag dimensions. In these cases, lower entropy (more concentrated feature importance) was an indicator of increased correctness. However, for this test, there were many insignificant results, including all of those relating to the XGBoost model. Therefore, this signal also cannot be relied on to always be an indicator of model failure, and must be assessed for each specific use case to confirm its presence.

Regarding hypothesis 3, despite some marginally significant results, overall there is no evidence to conclude that increased cross-method or cross-model agreement in XAI explanations correlates with prediction correctness.

Returning to the overarching question of whether local explanations can serve as a systematic discriminator between correct and incorrect predictions, the evidence is weak. Some signals – particularly regarding entropy – show some potential, and will be operationalised in Experiment 3, but in general local XAI explanations do not consistently serve as indicators of model failure. Any practitioners intending to use them as such must first validate that the signal actually exists for the specific use-case and method-model pair, and cannot assume it will be present.

Chapter 6

Exp. 3: XAI-Derived Trading Signals

This chapter presents Experiment 3, investigating whether integrating XAI-derived confidence signals can improve the performance of a simple trading strategy. The introduction states the objective of the third experiment: to test whether XAI outputs can be operationalised to improve trading performance, moving beyond post-hoc interpretation. It presents motivation and key prior literature for this specific experiment. The specific confidence gating strategy is then defined, and the results are presented and compared to the baselines.

6.1. Introduction

The potential for XAI explanations to serve as confidence signals in live trading strategies, beyond simple post-hoc analysis, was a core hypothesis of this thesis. Given that explanations provide insight into a model’s decision and the features driving them, and that prior literature has found shifts in attributions to indicate model failure, it is possible that systematic shifts in attributions could serve as a live signal of prediction confidence. If so, then filtering on that signal has the potential to improve trading performance. This would be extremely valuable for traders as a confidence filter that could be easily layered over any existing ML trading strategy, since it relies only on model-agnostic XAI methods.

The results of Experiment 2 demonstrated that it is difficult for XAI explanations to discriminate between correct and incorrect predictions, with no consistently viable discriminating signal. Even so, there were statistically significant discriminating signals that presented themselves for certain models, the strongest of which were: the Jensen-Shannon divergence of SHAP attributions for the LSTM’s lags, the entropy of LIME attributions for the LSTM’s lags, and the entropy of LIME attributions for the LSTM’s features. Therefore, these serve as strong candidate signals for this trading strategy test.

This experiment aims to evaluate trading strategies derived from XAI explanations, using these discriminating signals as confidence filters. They will be compared to buy-and-hold and all-predictions (no filter) benchmarks. The evaluation procedure includes a fully out-of-sample backtest over a 2-year period, and a comprehensive set of standard evaluation metrics for trading strategies. These include measures of returns, risk measures such as volatility and maximum drawdown, and full-picture measures of risk-adjusted returns using Sharpe and Calmar ratios. By conducting this analysis, this experiment will answer the core question of this thesis: can XAI explanations be successfully operationalised as trading signals?

The remainder of this experiment is organised as follows: Section 6.2 details the experimental design, including the backtesting methodology and the evaluative metrics used; Section 6.3 formally defines the trading strategies, comprising the three XAI-derived trading strategies and two benchmark strategies; Section 6.4 presents the trading performance results; and Section 6.5 discusses these results in the context of the overarching aims of the experiment.

6.2. Experiment Design

This experiment operationalises the discriminating attribution signals identified as statistically significant in Experiment 2 into confidence-gating trading strategies, and tests whether filtering on these signals improves risk-adjusted performance relative to an unfiltered benchmark and a buy-and-hold baseline.

Three confidence-gating strategies are constructed from the LSTM-based signals identified in Experiment 2 (SHAP-Lag-JSD, LIME-Feature-Entropy, LIME-Lag-Entropy). These strategies are detailed in Section 6.3. These strategies are evaluated against two benchmarks: an LSTM all-predictions (unfiltered) strategy, and a buy-and-hold baseline.

Backtesting is conducted on test set 2, an additional 2-year held-out test period (01/01/2024 – 31/12/2025) distinct from the data used in Experiments 1 and 2. This ensures the strategy evaluation is truly out-of-sample. It is assumed that there are no transaction costs, market impact, or slippage. Performance is evaluated using annualised returns, Sharpe ratios, Calmar ratios, maximum drawdowns, and volatilities, allowing comparison of both risk-adjusted return and downside risk across strategies. Exposure (the proportion of weeks traded) is also reported.

6.2.1. Annualised return

The annualised return is computed by compounding the mean weekly return over 52 periods:

$$R_{ann} = (1 + \bar{R}_{weekly})^{52} - 1,$$

where $\bar{R}_{weekly}$ is the mean weekly return over the backtest period.

6.2.2. Volatility

Strategy risk is measured directly via the standard deviation of strategy returns:

$$\sigma = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2},$$

computed over the backtest period and annualised by scaling by $\sqrt{52}$ (for a weekly frequency). This allows raw riskiness to be compared across strategies, alongside their risk-adjusted performance.

Volatility is also computed using only returns on the weeks the strategy actually trades (σ_{trades}). This is to account for the fact that, under the standard volatility measure, the filtered strategies would naturally have lower volatility due to the presence of 0-position weeks reducing variation in returns.

6.2.3. Sharpe ratio

Risk-adjusted return is measured using the Sharpe ratio (Sharpe, 1966), defined as:

$$Sharpe = \frac{\bar{R}}{\sigma},$$

where $\bar{R}$ is the mean strategy return over the backtest period and σ is the standard deviation of strategy returns. The ratio is annualised by scaling the mean and standard deviation by $\sqrt{52}$ (for a weekly frequency). Higher values indicate greater return earned per unit of volatility risk.

6.2.4. Maximum drawdown

Maximum drawdown is an indicator of risk that measures the largest observed decline from a historical peak in cumulative strategy value:

$$MDD = \min_{t \in T} \left[\frac{\max_{s < t} V_s - V_t}{\max_{s < t} V_s} \right],$$

where V_t is the cumulative strategy value at time t . Expressed as a percentage, it captures worst-case peak-to-trough loss over the backtest period, independent of when in the sample it occurs.

6.2.5. Calmar ratio

The Calmar ratio (Young, 1991) evaluates return relative to downside (drawdown) risk rather than total volatility:

$$Calmar = \frac{R_{ann}}{MDD}.$$

Higher values indicate stronger return generation relative to the worst peak-to-trough loss experienced.

6.3. Trading Strategies

This section defines all the trading strategies implemented in this experiment. 3 XAI-derived strategies are assessed with reference to 2 benchmark strategies. The trading framework can be defined by:

$$R_t = h(x_t) \cdot p(x_t) \cdot r_t,$$

where x_t is the vector of input features, r_t is the log return in the energy index over week t , $p(x_t)$ is the directional position function (where $p(x_t) = -1, 0$, or $+1$), $h(x_t)$ is the filtering function, and R_t is the resulting return for the strategy over week t . The functions $h(x_t)$ and $p(x_t)$ differ between the trading strategy.

R_t is computed across the full test set for each strategy, resulting in a strategy returns series that can be evaluated using the previously defined evaluation metrics.

6.3.1. Benchmark strategies

Two benchmark strategies are implemented in this study: a buy-and-hold strategy, and an LSTM all-predictions strategy (no filtering).

- For the buy-and-hold strategy, $p(x_t) = h(x_t) = +1$ for all t .
- For the LSTM all-predictions strategy, $p(x_t) = \text{sign}(f(x_t))$ for all t , where $f(x_t)$ is the predictor function of the LSTM model. $h(x_t) = +1$ for all t .

6.3.2. XAI-derived strategies

All three XAI-derived strategies are based on the LSTM model, so $p(x_t) = \text{sign}(f(x_t))$ for all t . However, the filtering functions differ for each:

- For the SHAP-Lag-JSD strategy, JSD is computed between the aggregated lag importance distribution for the correct predictions in test set 1 (G_{lags}) and the lag importance distribution for the current prediction t ($g_{\text{lags}}(x_t)$): $J(x_t) = \text{JSD}(G, g_{\text{lags}}(f, x_t))$. The strategy trades when JSD is below a threshold J_0 : $h(x_t) = \mathbf{1}[J(x_t) \leq J_0]$.
- For the LIME-Feature-Entropy strategy, entropy is computed for the feature importance distribution for the prediction at week t ($g_{\text{features}}(f, x_t)$): $E(x_t) = H(g_{\text{features}}(f, x_t))$. The strategy trades when entropy is below a threshold E_0 : $h(x_t) = \mathbf{1}[E(x_t) \leq E_0]$.
- The LIME-Lag-Entropy strategy operates identically, with $g_{\text{features}}(f, x_t)$ replaced by $g_{\text{lags}}(f, x_t)$, and a different threshold value E_0 .

All threshold values are determined by computing JSD/entropy values on test set 1, and selecting the threshold value that would maximise directional accuracy if implemented on that test set, with the additional constraint that between 50% and 80% of the test set 1 values must be below the threshold. The lower limit aims to provide sufficient exposure is for a reliable test, while the upper limit aims to ensure the strategy is meaningfully different from the all-predictions strategy (enough trades being filtered to draw a reliable conclusion beyond the random chance of filtering a few bad trades). Setting thresholds based on test set 1 ensures they are not fit on the same data used to evaluate trading performance, preserving the out-of-sample validity of the backtest.

6.4. Results

Table 6.1 presents the key out-of-sample metrics for the three XAI-derived trading strategies (based on the significant results from Experiment 2), as well as a buy-and-hold baseline, and an all-predictions baseline. Since the most significant Experiment 2 results

were all based on the LSTM model, the trading strategies all use this model, and only the LSTM's all-predictions strategy is needed as a baseline.

Table 6.1: Performance evaluation metrics of all strategies (two benchmark strategies and three XAI-derived strategies)

Strategy	Total Return	R_{ann}	σ	σ_{trades}	Sharpe Ratio	MDD	Calmar Ratio	Exposure
Buy-and-Hold	-0.192	-0.108	0.209	0.209	-0.546	0.250	-0.431	1.000
LSTM All-Preds	0.018	0.010	0.210	0.210	0.046	0.330	0.029	1.000
LSTM-SHAP-Lag-JSD	0.149	0.077	0.149	0.194	0.500	0.179	0.431	0.588
LSTM-LIME-Feature-Entropy	0.082	0.043	0.200	0.216	0.211	0.279	0.155	0.866
LSTM-LIME-Lag-Entropy	0.084	0.044	0.138	0.177	0.313	0.157	0.280	0.608

Note that the all-predictions strategy already delivers an improvement over the buy-and-hold strategy, with higher returns, a lower maximum drawdown, and higher Sharpe and Calmar ratios. This suggests the model has good predictive power in this additional out-of-sample test set too.

All three XAI-derived strategies show noticeable improvement over the all-predictions strategy, and large improvements over the buy-and-hold baseline. For all three, returns are higher, maximum drawdowns are lower, and Sharpe and Calmar ratios are higher than the baselines. Importantly, they do this while maintaining a reasonable level of exposure (>58%), so the result is not simply a consequence of executing a small number of trades. Volatility is also lower for the XAI-derived strategies, and this persists for two of the three strategies even when measuring volatility only over weeks when trades actually occurred.

The SHAP-lag-JSD strategy displays the best performance, with a Sharpe ratio of 0.500. The LIME-lag-entropy and LIME-feature-entropy strategies performed slightly weaker, with Sharpe ratios of 0.313 and 0.211 respectively. Although not particularly high in isolation, these results are meaningful given the buy-and-hold strategy made negative returns during this period and achieved a Sharpe ratio of -0.546.

Figure 6.1 displays the cumulative returns for each strategy. The standalone LSTM predictions allowed all the LSTM strategies to avoid two periods of consistently negative returns, where the buy-and-hold strategy suffered significantly. The SHAP-lag-JSD and LIME-lag-entropy strategies provided further gains by avoiding multiple periods of negative returns during Q1 and Q2 2025, where the LSTM all-predictions model suffered. The LIME-feature-entropy strategy more closely tracked the all-predictions strategy during these and other periods, instead benefitting from smaller on-off filtering decisions that allowed it to out-perform the all-predictions strategy. Overall, the

strategies demonstrated a significant ability to filter poor LSTM model predictions, providing additional gains over those of the raw LSTM model.

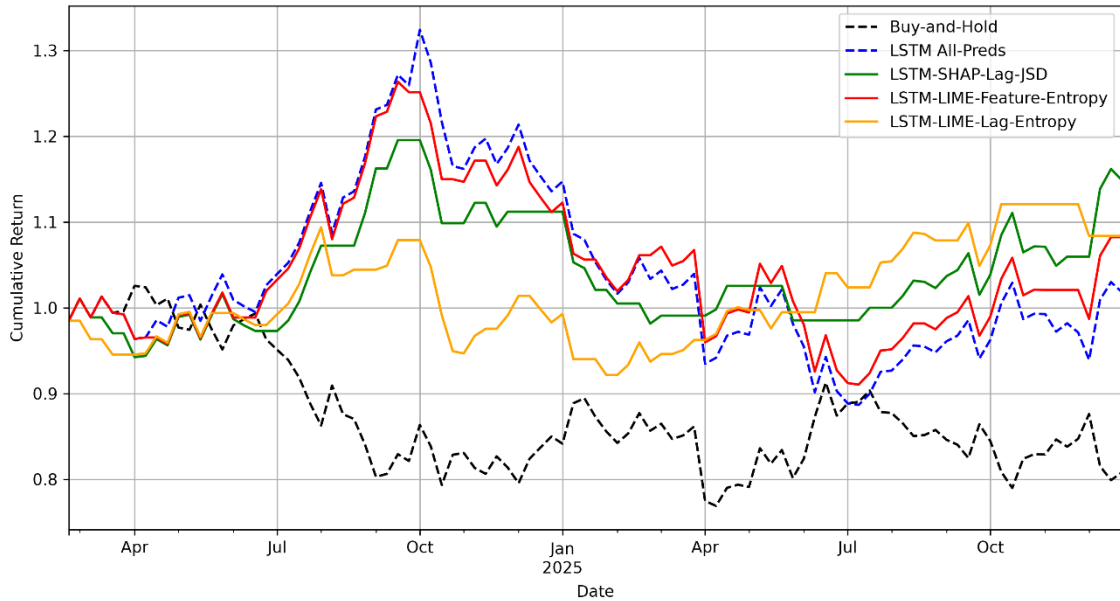


Figure 6.1 Cumulative returns of the XAI-derived trading strategies and benchmarks over the backtest period

However, the filtering also dampened the gains during the bullish period of Q2 2024, indicating that overall performance is dependent on the specific market environment. This is expected for any confidence-filtering strategy, as they are designed to be more cautious and so will perform worse during bullish regimes and better during bearish regimes. The key question is whether the benefit of correctly identifying incorrect predictions outweighs the cost of incorrectly excluding correct ones. Within this limited test period, the evidence suggests it does.

6.5. Discussion

This experiment implemented the statistically significant discriminating signals from Experiment 2 in a live trading strategy, operationalising XAI explanations as confidence signals to filter trading signals. Three strategies were built based on the LSTM model: SHAP-Lag-JSD, LIME-Feature-Entropy, LIME-Lag-Entropy. These were compared to two benchmarks: a buy-and-hold baseline and an LSTM all-predictions strategy.

The results show strong and consistent support for operationalising XAI explanations as confidence signals, with all three XAI-derived trading strategies outperforming both benchmarks across all metrics. The SHAP-lag-JSD strategy displayed the best performance, achieving a Sharpe ratio of +0.500, a significant 1.046 higher than the buy-and-hold strategy. The other strategies also delivered positive Sharpe ratios. They were able to – to differing extents – avoid the periods of poor prediction performance that the LSTM all-predictions strategy suffered from.

However, this filtering also had the effect of lower returns than the all-predictions model during extremely bullish market periods. Therefore, more testing is necessary over a longer time period and a wider range of market regimes, to assess whether these findings are generalisable.

Chapter 7

Conclusion and Future Work

This chapter starts by summarising the key findings of this thesis. It then makes clear the applications and contributions to science and to industry. Finally, it notes the limitations of this research, and suggestions for future work in this area.

7.1. Summary

This thesis has presented a broad analysis of how both established and recent XAI methods perform on a diverse set of energy index forecasting models, and introduced a novel mechanism to operationalise XAI explanations as confidence signals in live trading strategies. These findings will aid practitioners in applying XAI methods appropriately and offer potential ways to improve trading performance of existing ML strategies.

This thesis showed that while different XAI methods tend to converge on the major patterns in a model's decision-making, they can also differ significantly, and can sometimes produce degenerate results. It has also demonstrated the usefulness of XAI evaluation metrics like faithfulness and stability for assessing whether explanations are trustworthy, and shown that faithfulness and stability can vary significantly between XAI methods and depend on the forecasting models they are applied to. In particular, both SHAP and LIME performed worse on the LSTM model than the non-temporal models, likely due to the higher feature count and autocorrelation present in the sequential input data, with LIME's feature importances being completely disconnected from reality. Notably, XAI evaluation results may differ from what other research has found on different datasets and so, while SHAP was the most consistently faithful and often the most stable in this analysis, it is demonstrated that evaluation of XAI performance in each specific use-case is essential.

The second and third experiments demonstrated a novel way to operationalise XAI explanations as confidence signals, filtering a trading strategy to improve risk-adjusted returns. At the first stage, this involved a rigorous examination of potential statistical signals to discriminate between correct and incorrect model predictions, including Jensen-Shannon divergence of feature importance distributions, entropy of attributions, and cross-method agreement. Results indicated that while discriminating signals can sometimes exist, particularly for the LSTM and for LIME, they are not guaranteed and not always consistent. The thesis then operationalised the three most promising discriminating signals in trading strategies and demonstrated consistent success when utilising these as confidence gates, with all three tested strategies outperforming both a buy-and-hold and an all-predictions benchmark in an out-of-sample backtest.

These results provide evidence in support of the central hypothesis of this thesis – that XAI explanations can be successfully operationalised in trading strategies to improve risk-adjusted performance – with the important caveat that this is dependent on a discriminating signal being present, which is not guaranteed for every model-XAI method combination even on a single dataset.

7.2. Contributions

This section presents the key contributions of this thesis.

Firstly, this thesis demonstrated the importance of testing and comparing multiple XAI methods rather than implementing one and assuming its explanations are trustworthy, with Experiment 1 surfacing a highly degenerate set of explanations produced by LIME on the LSTM model. This is essential guidance for future research that implements XAI analysis.

Secondly, the cross-method, cross-model faithfulness and stability results showed that SHAP was consistently the most faithful and often the most stable method, providing confidence in its use as the most popular XAI method and indicating to academics that they should include SHAP within their primary toolset.

Thirdly, Experiment 1 provided guidance on how to evaluate XAI methods using measures of faithfulness and stability. It also demonstrated weaknesses of these standard evaluation metrics, emphasising that they must be interpreted in context. LIME's deceptively strong Local Lipschitz stability on the LSTM was misleading, with the near-zero faithfulness indicating it failed to track the model rather than producing reliable but different explanations. Meanwhile, WinIT's near-zero faithfulness was attributable to a mismatch between its marginal-importance design and what occlusion-based metrics measure, and not to a flaw in the method itself.

Fourthly, this thesis demonstrated that Jensen-Shannon divergence and entropy of XAI attributions can serve to systematically discriminate between correct and incorrect predictions, although this is highly dependent on the model and method. Greater success was found for LIME and the LSTM model. This introduces a novel way to use XAI explanations beyond just post-hoc analysis.

Finally, this thesis provided evidence that XAI explanations can be operationalised as confidence filters in a live trading strategy, again with the important caveat that its benefit is contingent on a validated discriminating signal existing for the specific model in production, not something that can be assumed by default. The study presented a novel way to incorporate XAI explanations in a trading strategy, beyond simple post-hoc analysis, to improve risk-adjusted performance. This provides a new instrument that researchers, practitioners, and financial market participants can utilise to potentially improve trading performance, via a methodology that can be easily layered over any existing ML trading strategy.

The thesis also demonstrated that frontier models like TabPFN struggle to compete with the best traditional ML and DL models such as XGBoost and LSTM models on this weekly financial forecasting task, despite previous research demonstrating excellent performance on standard benchmark datasets. This provides guidance for practitioners on when it is appropriate to use these more advanced models, and suggests researchers should continue improving such models for a wider set of use cases.

7.3. Limitations

While the results of this thesis delivered many promising contributions, there are limitations to this study. One is the constrained sample size by the standards of backtesting research. The weekly frequency means that even though 22 years of data are used, this results in a limited number of observations. The training set sample size is much smaller than would be ideal, perhaps dampening the out-of-sample performance of the forecasting models. Similarly, test set 1 has limited observations, reducing the generalisability of the discriminating signals identified. It is also possible that genuine discriminating signals were missed due to them failing to present as statistically significant at this sample size. Using a daily dataset would help alleviate this issue, but would require a different set of features with a higher frequency.

However, test set 2's sample is particularly weak and this would not be fully resolved by using daily data, since it includes only 2 years of data. The standard deviation on an annualised Sharpe ratio over just 2 years is wide enough that the relative ranking of the trading strategies should be taken as indicative rather than precise. The short test period also limits the regime coverage. The results of Experiment 3 indicated that relative performance is dependent on regime type, highlighting the significance of this limitation.

Beyond sample size, it is important to note the scope of this study. The experiments are conducted on a single asset within a single asset class (the BCOMENTR energy index). It remains an open question whether the findings generalise to other assets and asset classes.

Model scope should also be noted. An intentionally diverse model stack was used in this study, encompassing a broad range of model paradigms: a tree-based model, a sequential neural-network model, and a transformer-based pre-trained foundation model. However, it is unproven whether these results also apply to other model types.

Finally, a key limitation of Experiment 3 was the assumption of no transaction costs, market impact, or slippage. A more realistic experiment could be designed to include these factors, and assess whether they impact the overall findings. The confidence-gating mechanism is also implemented as a binary threshold (trade or abstain) rather than a continuous confidence-weighted position size, which is a simplification relative to how such a signal could be deployed in practice.

7.4. Future work

There are several potential extensions that should be the focus of future research. Some of them address the previously discussed limitations, while others are promising directions for further investigation.

Regarding the sample size limitations, using higher frequency data would dramatically increase the number of observations, improving training so models are better able to extract signals from the data and are more likely to generalise well onto out-of-sample data, and improving the reliability and statistical power of analysis on the testing datasets. Using daily data would provide roughly 5x the number of observations. However, this would not alleviate all the sample size concerns, since the test sets will still cover the same time periods and limited regime types, so using a larger date-range would also be a valuable improvement in future research.

Testing on a broader range of assets, and across multiple asset classes would help assess whether the findings in this thesis can generalise across the wider financial markets, and provide stronger guidance on when practitioners can reliably implement the confidence signals proposed in this thesis. Similarly, testing on a wider range of model types would be valuable since trading strategies across the field utilise a wide variety of ML models. In particular, more recent and advanced model types should be investigated, such as temporal-fusion-transformers and agentic models, if XAI methods can be adapted to work on them.

A final avenue for future research would be to implement a more advanced backtesting framework for the trading strategies, incorporating transaction costs related to changes in positions. This would give a more accurate perspective on the profitability and performance of the strategies. More complex strategies could also be an interesting direction for future work, such as incorporating confidence signals for position sizing (rather than the simple long/short/no-position strategy filters used in this study), or using the confidence signals to drive model switching. The performance of these more complex strategies would be relevant to market participants interested in incorporating XAI-derived signals into their strategies.

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Appendices

Appendix A: Code and Data

The full code and data used in this thesis is available on request by emailing: kushal.patel.21@ucl.ac.uk.

It is also available at: <https://github.com/kush7000/COMP0077-Public>.

Appendix B: Summary statistics for the full feature set

Figure B.1: Summary statistics for all features considered prior to feature selection

Bloomberg ID	Description	Mean	Standard deviation	Min	Max	Start date
BCOMENTR Index	Energy commodities price index	178.0	170.2	17.9	919.5	1984-01-13
VIX Index	VIX Index	19.4	7.7	9.1	82.7	1990-01-02
OVX Index	OVX Index	38.7	17.4	14.5	325.2	2007-05-10
MOVE Index	MOVE Index	93.3	27.4	36.6	264.6	1988-04-04
DXY Curncy	USD FX	94.5	12.7	71.3	164.7	1985-01-01
FDTR Index	Fed Rate Upper Target	3.5	2.7	0.2	9.8	1985-01-01
FDFD Index	Fed Rate	2.5	2.2	0.0	9.0	1995-02-17
GT10 Govt	10Y Treasury Yield	4.8	2.4	0.5	12.0	1985-01-01
GDP CQOQ Index	US QoQ GDP Growth	2.7	4.2	-28.0	34.9	1985-01-01
BENWUSGC Index	US QoQ GDP Growth Nowcast	2.2	9.4	-30.5	32.0	2020-03-31
GDGCAFJP Index	US QoQ GDP Growth Nowcast	2.5	6.9	-53.8	37.0	2011-08-25
STLNENIC Index	US QoQ GDP Growth Nowcast	1.8	8.0	-49.8	20.6	2017-02-17
NAPMPMI Index	US Purchasing Managers Index	52.2	4.8	34.5	63.8	1985-01-01
SPX Index	S&P 500	1666.0	1502.7	163.7	6978.6	1985-01-01
ES1 Index	S&P 500 Futures	2229.0	1498.1	676.0	7016.5	1997-09-09
MXWO Index	MSCI World Index	1380.8	916.1	184.6	4578.1	1985-01-01
CL1 Comdty	WTI Price	47.8	29.2	-37.6	145.3	1985-01-01
CO1 Comdty	Brent Price	52.2	32.6	9.6	146.1	1988-06-23
HOCL1 Index	Heating Oil Spread	25.4	12.6	3.2	97.2	2009-07-02
USCSMEHC Index	Cushing Crude Price	0.8	1.8	-8.6	14.8	2017-01-23
CRK321M1 Index	321 Crack Spread	12.4	9.9	1.0	61.0	1986-11-25
HUCL1 Index	Gasoline Spread	24.4	354.1	-2.7	23443.7	2009-07-02
CLCO1 Index	WTI-Brent Spread	-4.1	5.6	-27.9	6.4	2003-04-23
DSUPKERO Index	Kerosene Days of Supply	26.9	4.4	20.0	73.2	1991-03-08
DSUPGASL Index	Gasoline Days of Supply	25.3	2.8	19.8	48.7	1991-03-08
DOEICRP2 Index	Cushing Crude Imports	1562.1	845.3	311.0	3441.0	1990-01-05
BAKETOT Index	Active US Oil & Gas Rigs	1054.1	456.8	244.0	2713.0	1985-01-01
BAKEGAS Index	Active US Gas Rigs	530.8	394.2	68.0	1606.0	1987-07-17
DOESCRUD Index	US Crude Inventory	349178.1	67825.5	247323.0	540722.0	1985-01-01
DOESCROK Index	Cushing Crude Inventory	35169.5	14419.2	11677.0	69420.0	2004-04-09
DSUPCRUD Index	Crude Days of Supply	23.6	4.0	16.1	42.0	1985-01-01
DOEPGRIN Index	US Crude Processed Volume	15317.9	1167.1	10300.0	18243.0	1989-09-08
DOEPPERC Index	US Refinery Utilization Rate	89.6	4.9	56.0	100.5	1989-09-08
DSUPDIST Index	Distillate Days of Supply	34.6	6.0	21.8	54.3	1991-03-08
EUR Curncy	EUR FX	1.2	0.2	0.6	1.6	1985-01-01
IP Index	US Industrial Production	86.4	16.1	54.5	104.1	1985-01-01
CAD Curncy	CAD FX	1.3	0.2	0.9	1.6	1985-01-01
INJCJC Index	US Jobless Claims	363.1	276.5	190.0	6137.0	1985-01-01
CNVSPSGR Index	China Automobile Sales	1502894.6	694911.7	213534.0	3120031.0	2005-01-31
EPUSTOT Index	US Electricity Output	71999.0	10685.1	49132.0	99445.0	1989-01-06

Appendix C: Stationarity test results and feature transformations

Figure C.1: P-values from ADF and KPSS tests on candidate features, and the transformation applied to account for non-stationarity

Feature	ADF test p-value	KPSS test p-value	Transformation applied
VIX Index	0.000	0.100	No transformation
MOVE Index	0.000	0.010	No transformation
DXY Curncy	0.000	0.010	No transformation
FDTR Index	0.129	0.010	Difference
FDFD Index	0.171	0.010	Difference
GT10 Govt	0.065	0.010	Log-returns
NAPMPMI Index	0.001	0.100	No transformation
SPX Index	1.000	0.010	Log-returns
ES1 Index	0.999	0.010	Log-returns
MXWO Index	0.998	0.010	Log-returns
CL1 Comdty	0.364	0.010	Log-returns
CO1 Comdty	0.367	0.010	Log-returns
CRK321M1 Index	0.318	0.010	Difference
CLCO1 Index	0.144	0.010	Difference
DSUPKERO Index	0.000	0.100	No transformation
DSUPGASL Index	0.000	0.010	No transformation
DOEICRP2 Index	0.965	0.010	Difference
BAKETOT Index	0.046	0.010	Difference
BAKEGAS Index	0.472	0.010	Difference
DOESCRUD Index	0.590	0.010	Difference
DOESCROK Index	0.081	0.010	Difference
DSUPCRUD Index	0.076	0.010	Difference
DOEPGRIN Index	0.003	0.010	Difference
DOEPPEREC Index	0.000	0.020	No transformation
DSUPDIST Index	0.000	0.089	No transformation
EUR Curncy	0.005	0.035	No transformation
IP Index	0.386	0.010	Difference
CAD Curncy	0.391	0.010	Difference
INJCJC Index	0.000	0.100	No transformation
EPUSTOT Index	0.010	0.010	Difference
CL1 H/L	0.000	0.100	No transformation